

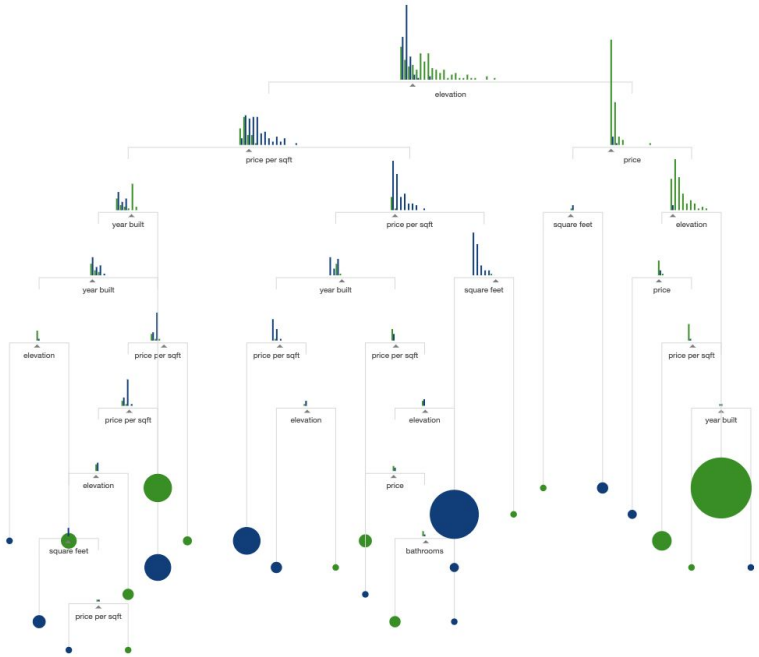
JKU
Visual Data
Science Lab

AI and Visualization

Explaining Through Projections

Univ.-Prof. Dr. Marc Streit

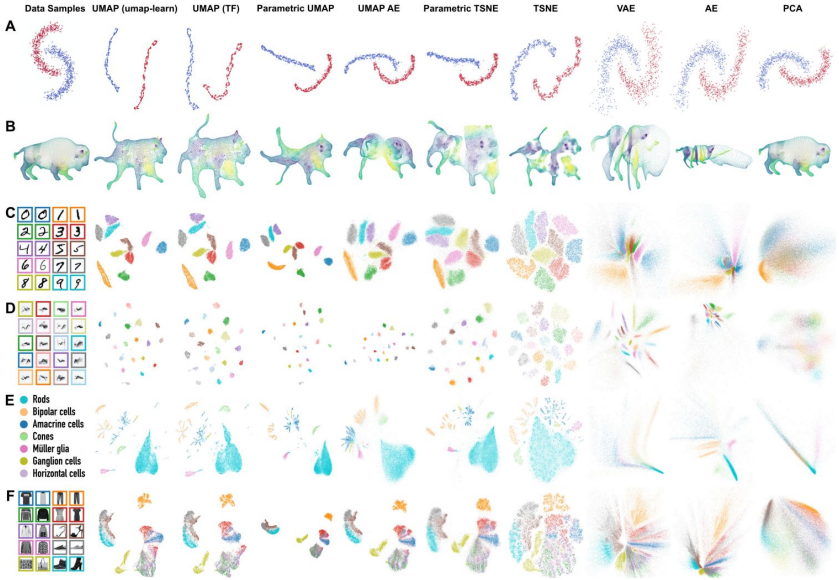
ai-and-vis-course@jku.at



Recap of Lecture 2

Fundamentals and Explaining Algorithms

- **AI & VIS Fundamentals**
- **Explaining**
  - Sorting Algorithms
  - Clustering Algorithms & Results
  - Bayes' Theorem
  - Decision Trees

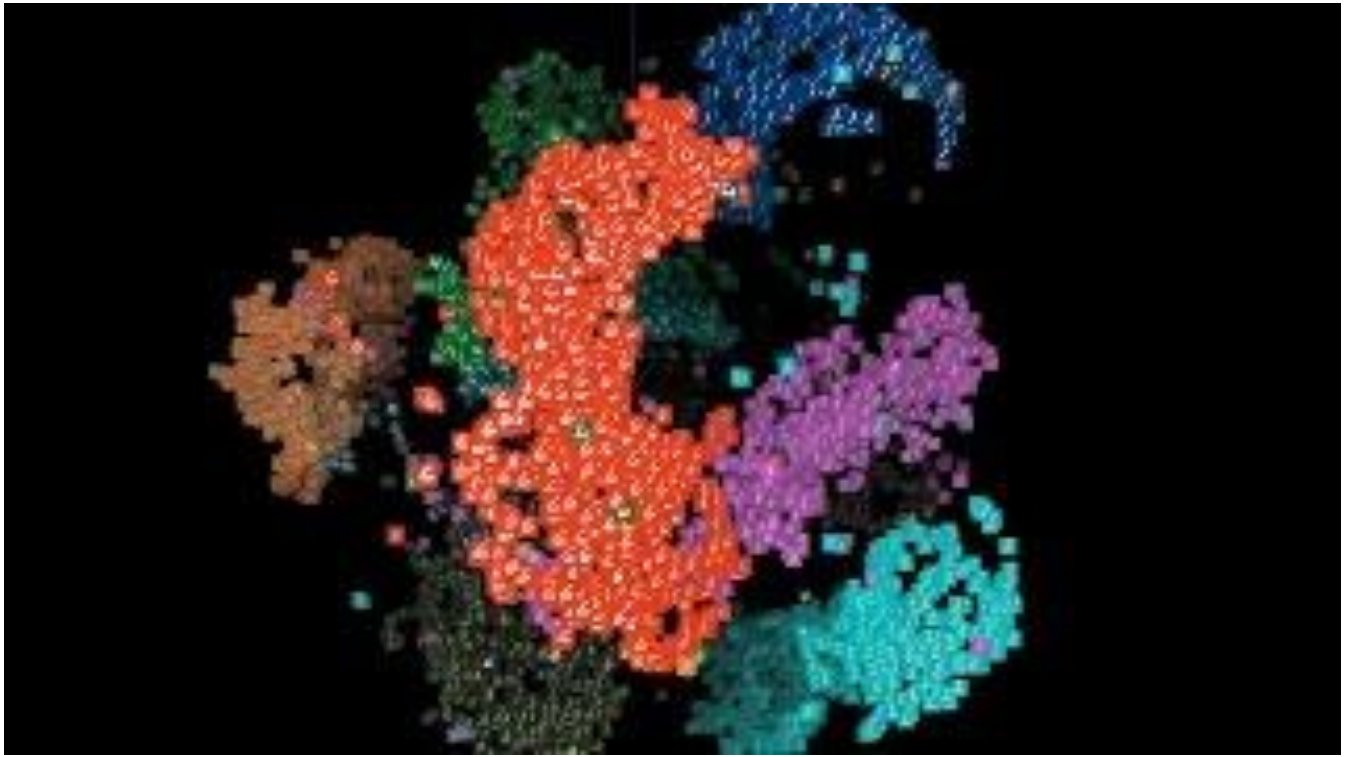


Lecture 3

Explaining Through Projections

- **Dimensionality reduction techniques**
  - PCA, UMAP, t-SNE
  - Visualization of embeddings
- **Embedding-based trajectories**

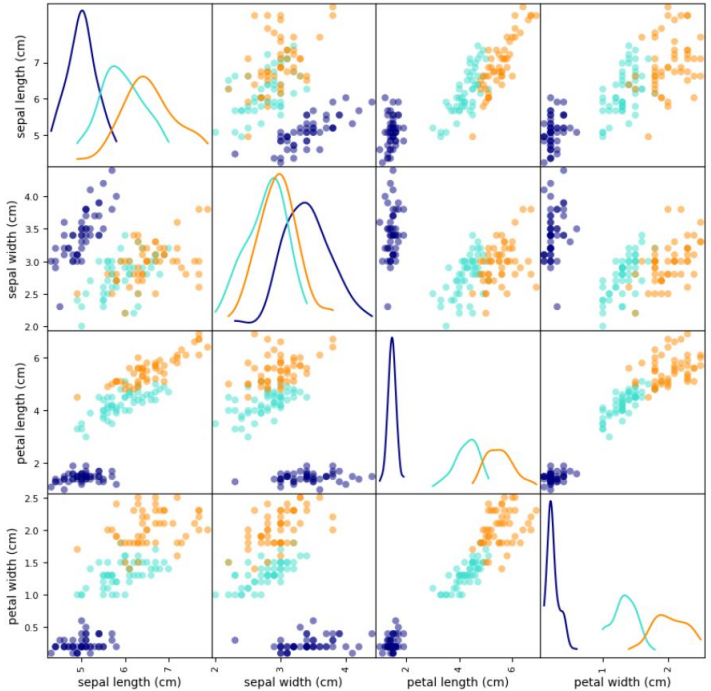
Introduction by Viegas and Wattenberg from Google on Visualizing High-Dimensional Spaces

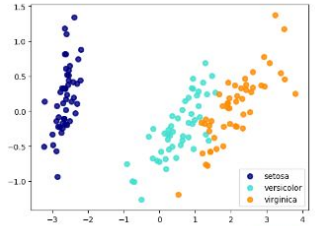


Low-Dimensional Embeddings / Projections

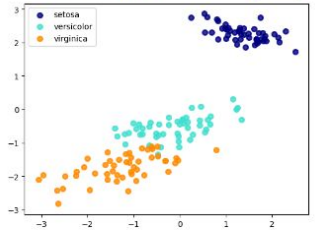


Iris
Dataset

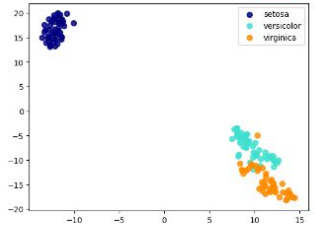




(b) Dataset projection with PCA



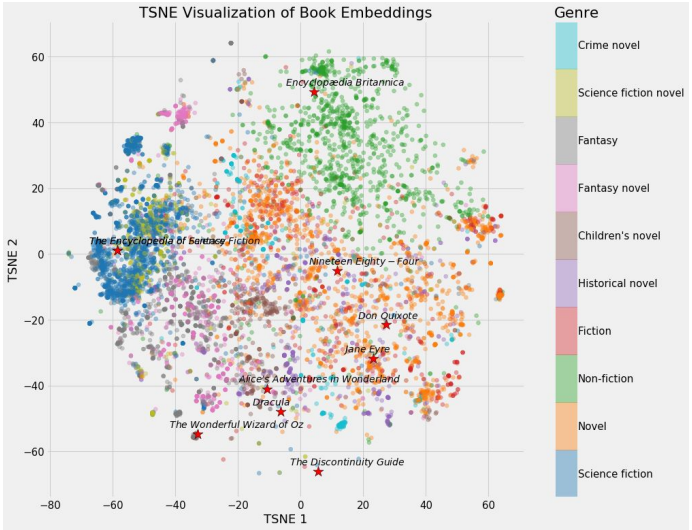
(c) Dataset projection with MDS



(d) Dataset projection with t-SNE

Dimensionality Reduction

- Curse of dimensionality in many real world problems
- Transform high-dim data to space with fewer dimensions (2D/3D)
- Powerful technique to look for (hidden) structure in high-dimensional data
- Used in various domains
  - Document categorization, drug discovery, machine learning model debugging, etc.
- Disadvantages
  - Hard to preserve semantics of single dimensions
  - Hard to understand and interpret
  - Error not visible —> inspiring false confidence



Embedding of all books on Wikipedia

Projecting Data from nD to (1D)/2D/3D

- Goal
  - Reveal patterns and clusters of similar or dissimilar data
- Different terms
  - Dimensionality reduction, (down)projection, embedding
- Various DR techniques/algorithms exists
  - All of them have different strengths & weaknesses

- Data items described by high-dimensional feature vector

The Beginner's Guide to Dim. Reduction

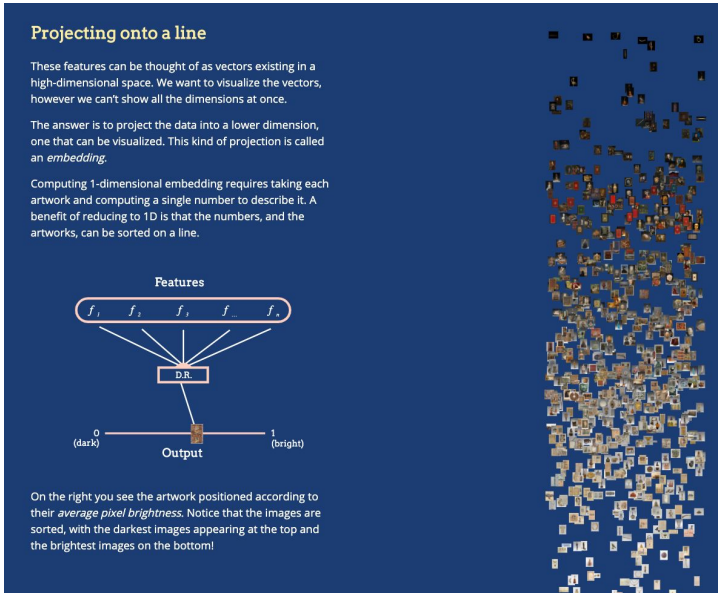
Matthew Conlen and Fred Hohman, July 2018

- Nice scrollytelling intro
- Dataset
  - 800 artworks from the collection at the Metropolitan Museum of Art

Year	Title	Artist
1680	Spindle-back armc...	
1875	Armchair	Pottier and Stymus ...
1776	Basket	Myer Myers
1650	Basin	Master Potter A
1710	Two-handled Bowl	Cornelius Kierstede



Clothing the Naked (1666)
Michiel Sweerts



https://dimensionality-reduction-293e465c2a3443e8941b016d.vercel.app/

For the mathematically inclined

Dimensionality reduction can be formulated mathematically in the context of a given dataset. Consider a dataset represented as a matrix X, where X is of size $m \times n$, where m represents the number of rows of X, and n representes the number of columns.

Typically, the rows of the matrix are *data points* and the columns are *features*. Dimensionality reduction will reduce the number of features of each data point, turning X into a new matrix, X', of size $m \times d$, where $d < n$. For visualizations we typically set d to be 1, 2 or 3.

Say $m = n$, that is X is a square matrix. Performing dimensionality reduction on X and setting $d = 2$ will change it from a square matrix to a tall, rectangular matrix.

$$X = \begin{bmatrix} x & x & x \\ x & x & x \\ x & x & x \end{bmatrix} \implies \begin{bmatrix} x' & x' \\ x' & x' \\ x' & x' \end{bmatrix} = X'$$

Each data point only has two features now, i.e., each data point has been reduced from a 3 dimensional vector to a 2 dimensional vector.

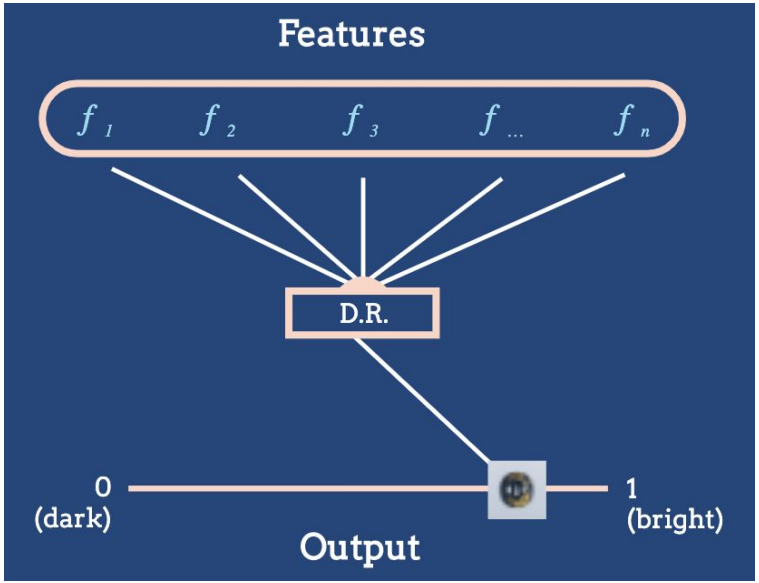
Dataset can consist of:

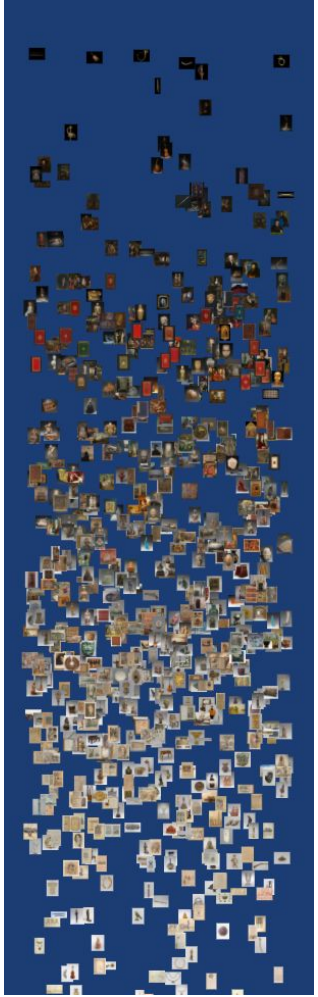
- Images
- Words
- Some other high-dim. vectors

https://dimensionality-reduction-293e465c2a3443e8941b016d.vercel.app/

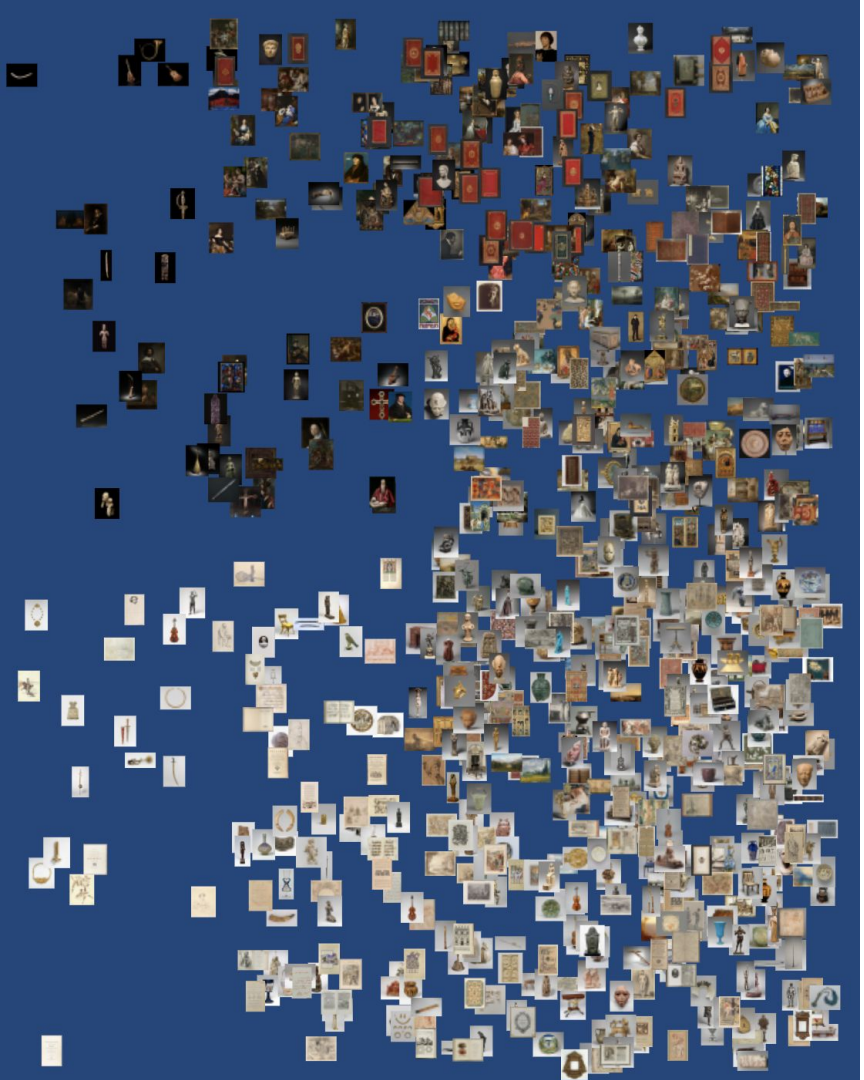
Projecting to 1D Space

Artwork positioned according to their average pixel brightness.





https://dimensionality-reduction-293e465c2a3443e8941b016d.vercel.app/



Projecting to 2D Space

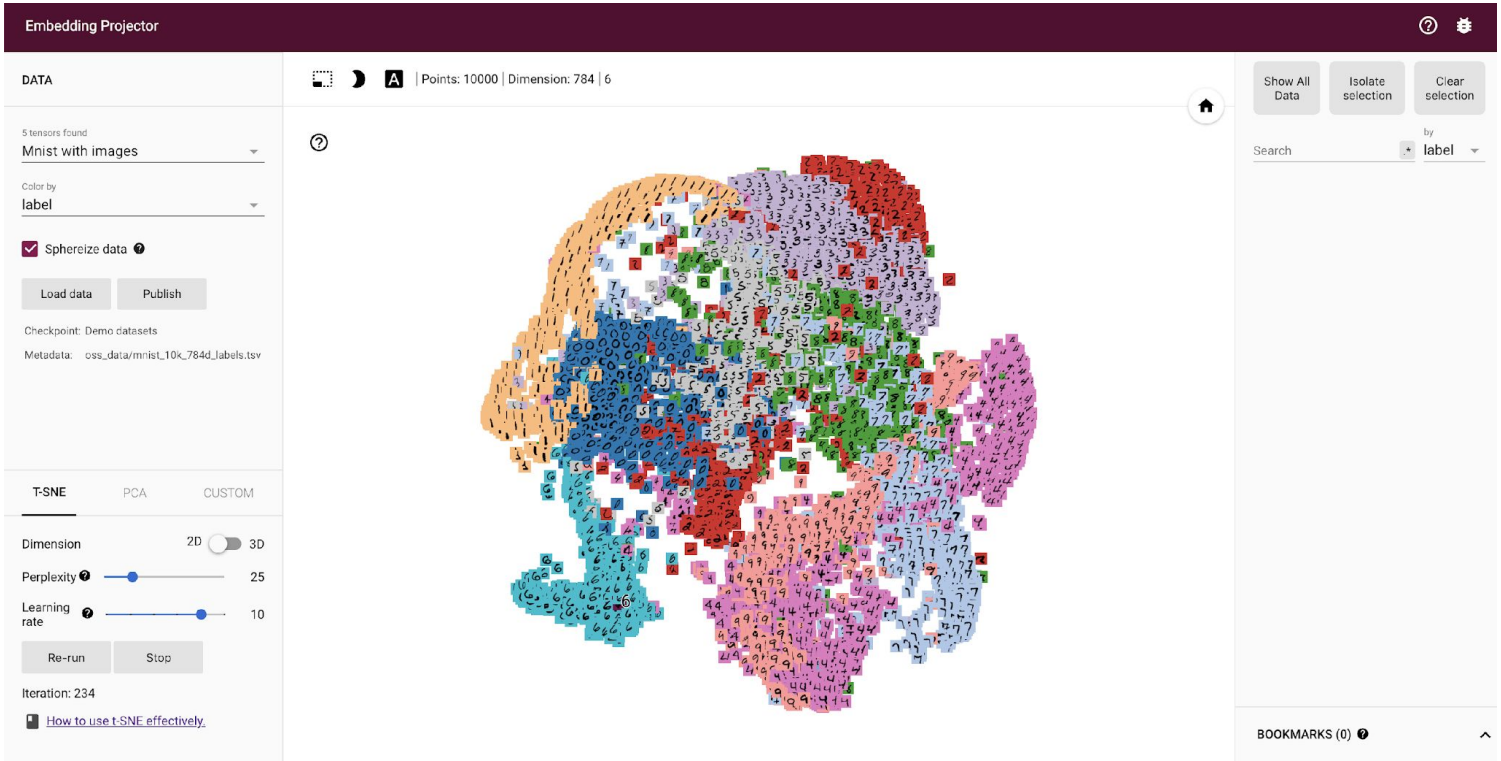
Based on image brightness

The pieces have more room to spread out.

https://dimensionality-reduction-293e465c2a3443e8941b016d.vercel.app/

Play Around with Google Embedding Projector

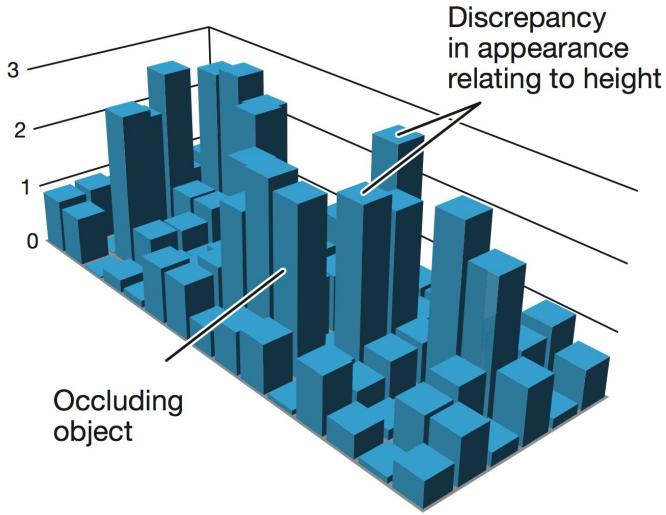
https://projector.tensorflow.org/

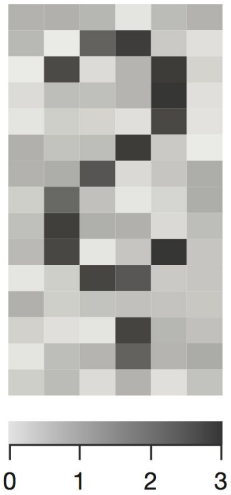


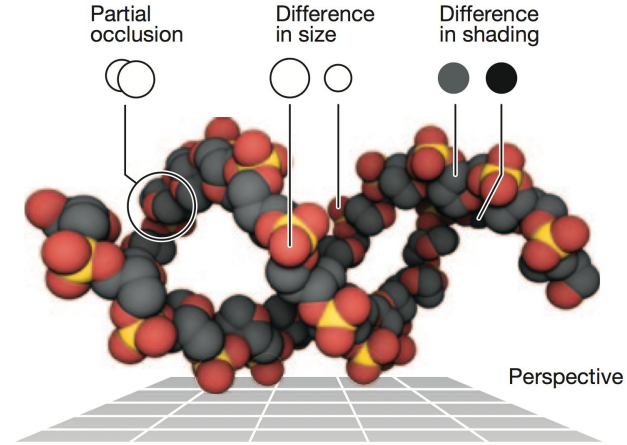
Vis Excursion: Be Careful with 3D!

Which one is the tallest bar? What is the pattern in the data?

Rule: No unjustified 3D! Exception: 3D Phenomena



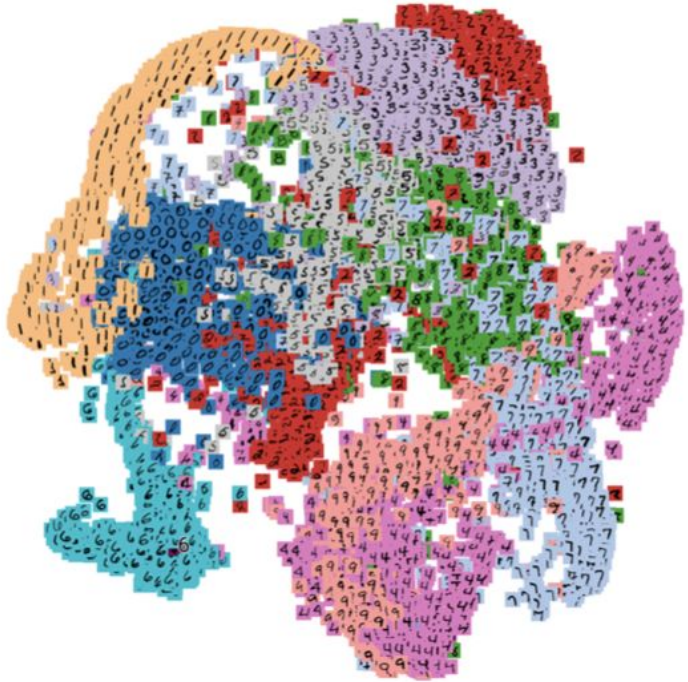




Depth cues enable us to perceive 2D images as 3D objects
Benefit: Shape perception

Representing Data Points / Items in Projections

- Categorical attributes / features
  - Hue
  - Shape
  - e.g., for class assignment
- Ordered attributes / features
  - Size
  - Brightness / saturation
- Some other item representation
  - Image
  - Glyph



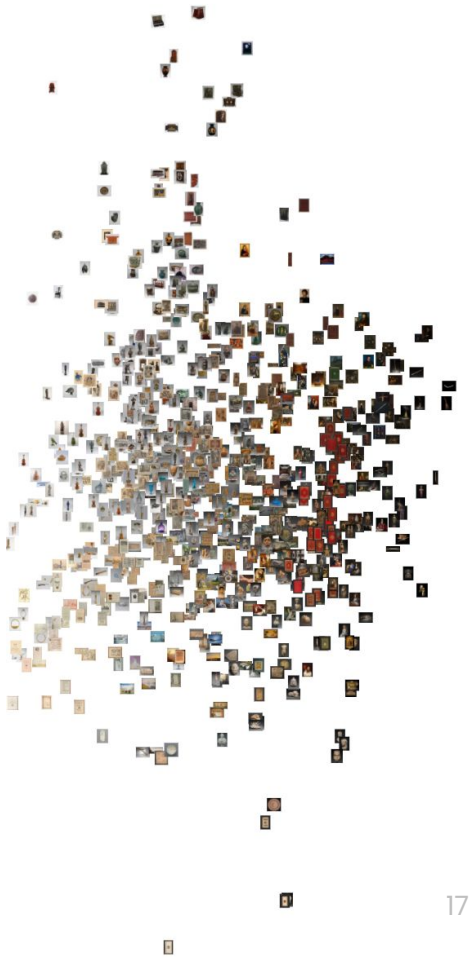
Dimensionality Reduction Techniques

Dimensionality Reduction Techniques

- Linear approaches
  - Principle Component Analysis (PCA)
  - Multidimensional Scaling (MDS)
- Non-linear approaches
  - t-distributed stochastic neighbor embedding (t-SNE)
  - Uniform manifold approximation and projection (UMAP)
  - Self-Organizing Maps (SOM)
- For more details and how they check unsupervised learning courses

Principal Component Analysis (PCA)

- Pros
  - Relatively computationally cheap.
  - Can save embedding model to then project new data points into the reduced space.
- Cons
  - Linear reduction limits information that can be captured; not as discriminably clustered as other algorithms.



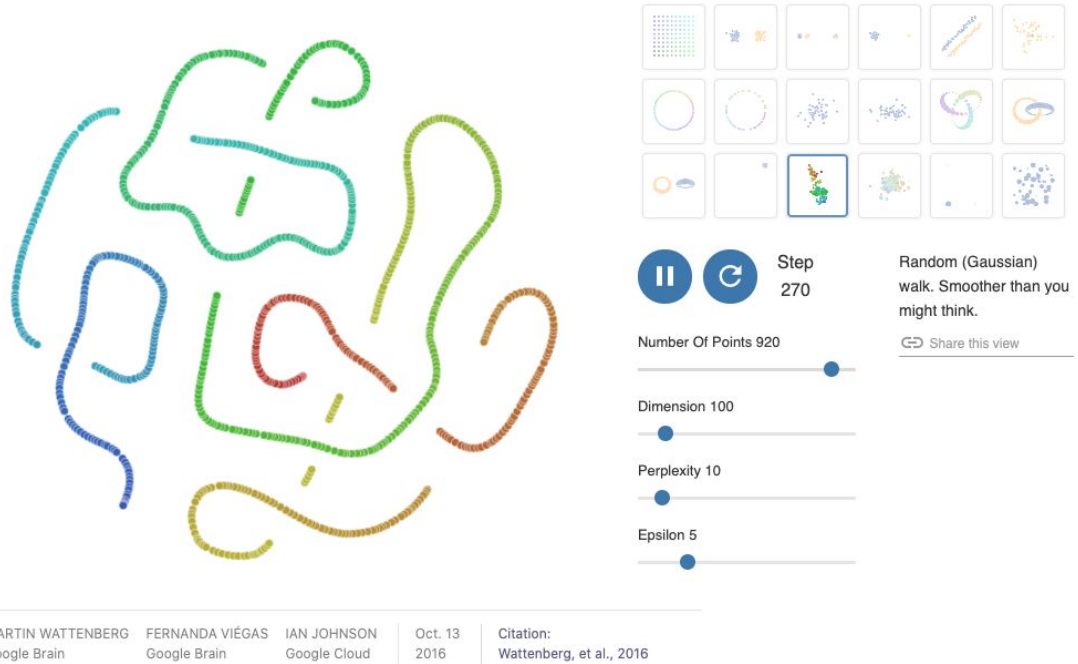
t-Distributed Stochastic Neighbor Embedding (t-SNE)

- Pros
  - Produces highly clustered, visually striking embeddings.
  - Non-linear reduction, captures local structure well.
- Cons
  - Global structure may be lost in favor of preserving local distances.
  - More computationally expensive.
  - Requires setting hyperparameters that influence quality of the embedding.
  - Non-deterministic algorithm.



Distill Article: How to Use t-SNE Effectively

https://distill.pub/2016/misread-tsne/



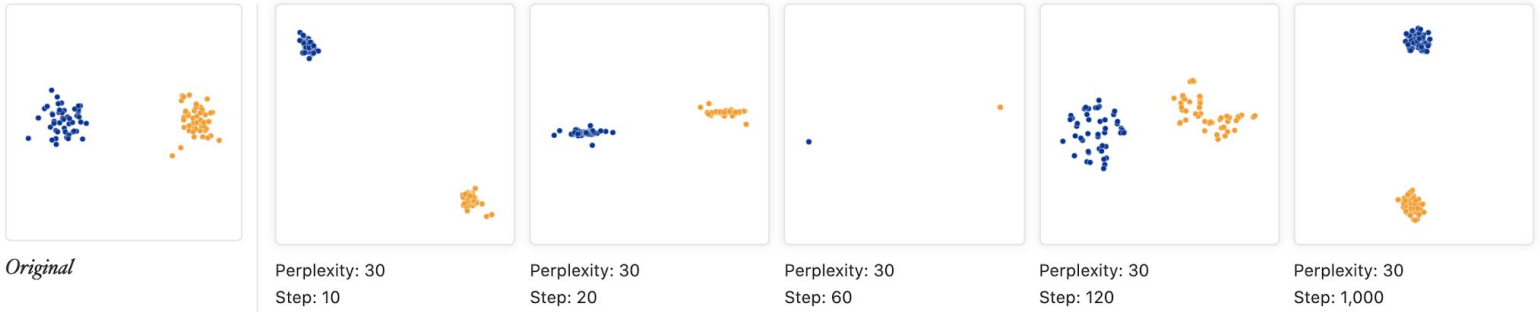


BE CAREFUL WHEN INTERPRETING PATTERNS!

1. Those hyperparameters really matter



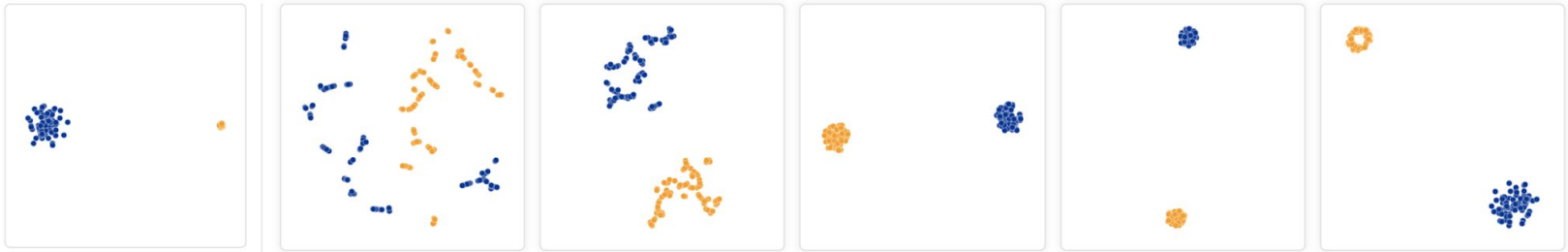
t-SNE plots for five different perplexity values



Five different runs at perplexity 30. The first four were stopped before stability.

https://distill.pub/2016/misread-tsne/

2. Cluster sizes in a t-SNE plot mean nothing



Original

Perplexity: 2
Step: 5,000

Perplexity: 5
Step: 5,000

Perplexity: 30
Step: 5,000

Perplexity: 50
Step: 5,000

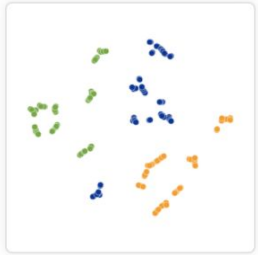
Perplexity: 100
Step: 5,000

https://distill.pub/2016/misread-tsne/

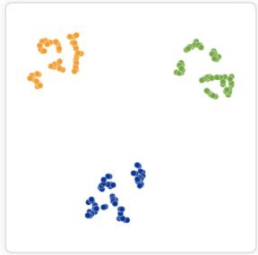
3. Distances between clusters might not mean anything



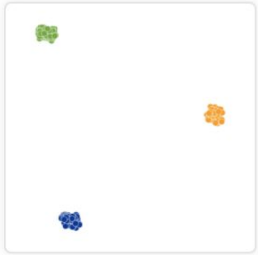
Original



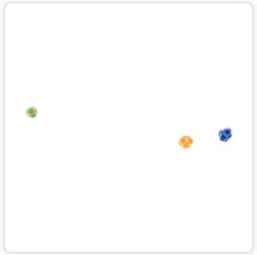
Perplexity: 2
Step: 5,000



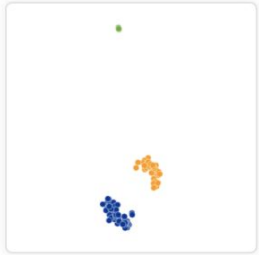
Perplexity: 5
Step: 5,000



Perplexity: 30
Step: 5,000



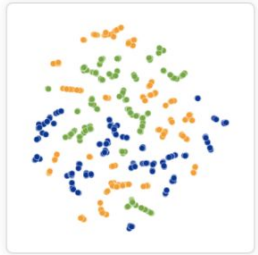
Perplexity: 50
Step: 5,000



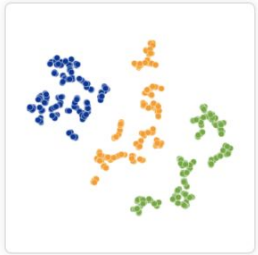
Perplexity: 100
Step: 5,000



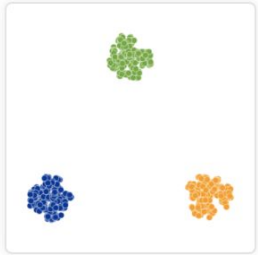
Original



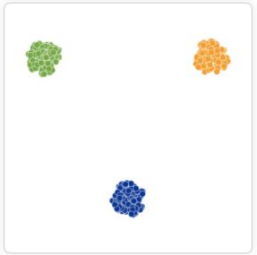
Perplexity: 2
Step: 5,000



Perplexity: 5
Step: 5,000



Perplexity: 30
Step: 5,000



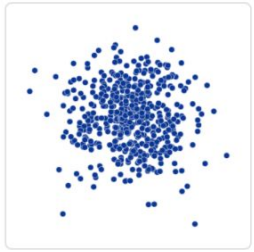
Perplexity: 50
Step: 5,000



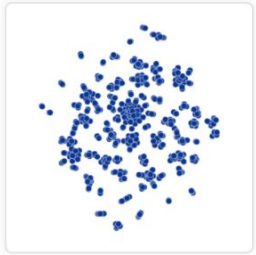
Perplexity: 100
Step: 5,000

https://distill.pub/2016/misread-tsne/

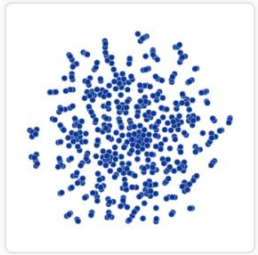
4. Random noise doesn’t always look random



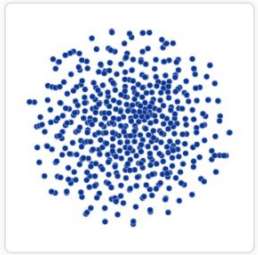
Original



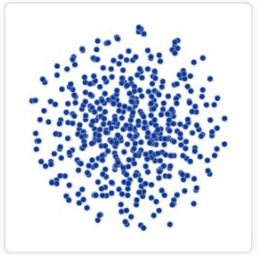
Perplexity: 2
Step: 5,000



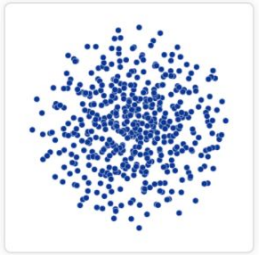
Perplexity: 5
Step: 5,000



Perplexity: 30
Step: 5,000



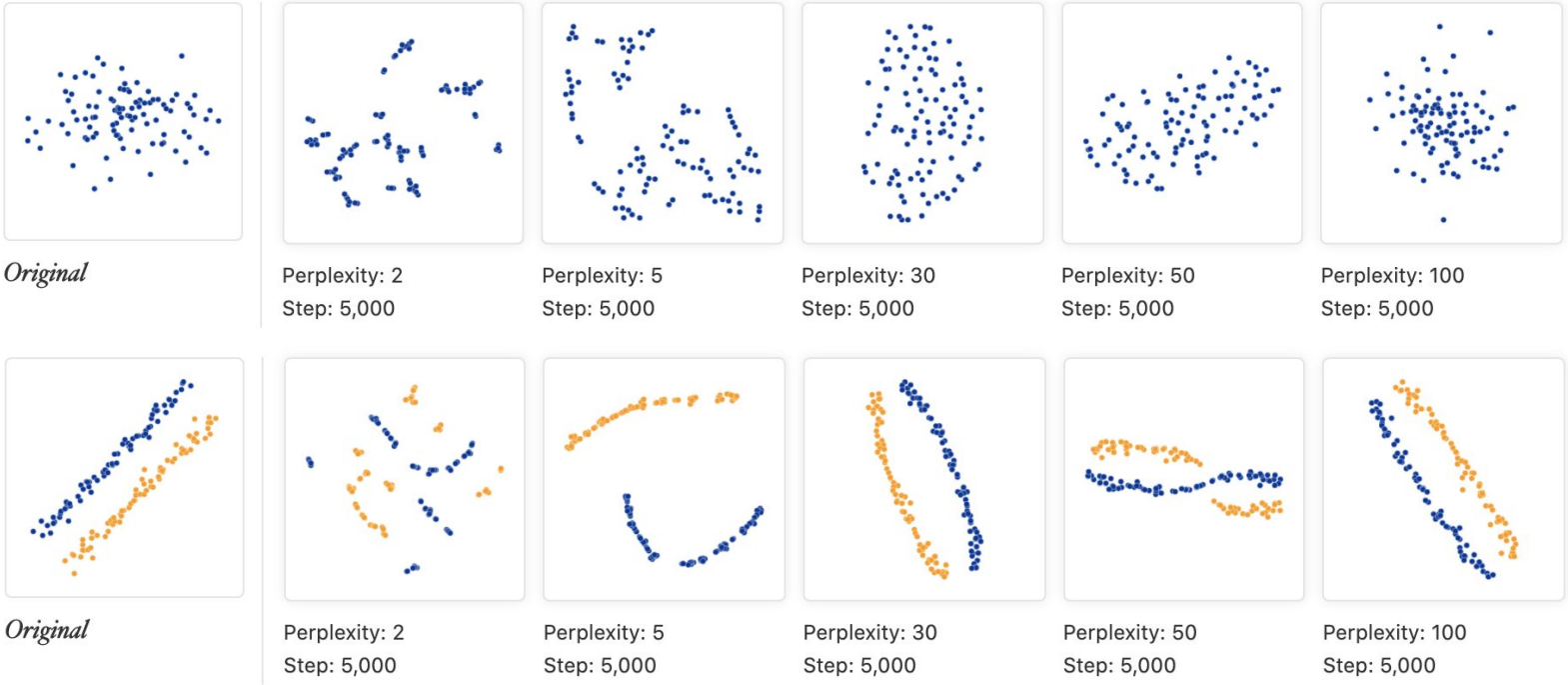
Perplexity: 50
Step: 5,000



Perplexity: 100
Step: 5,000

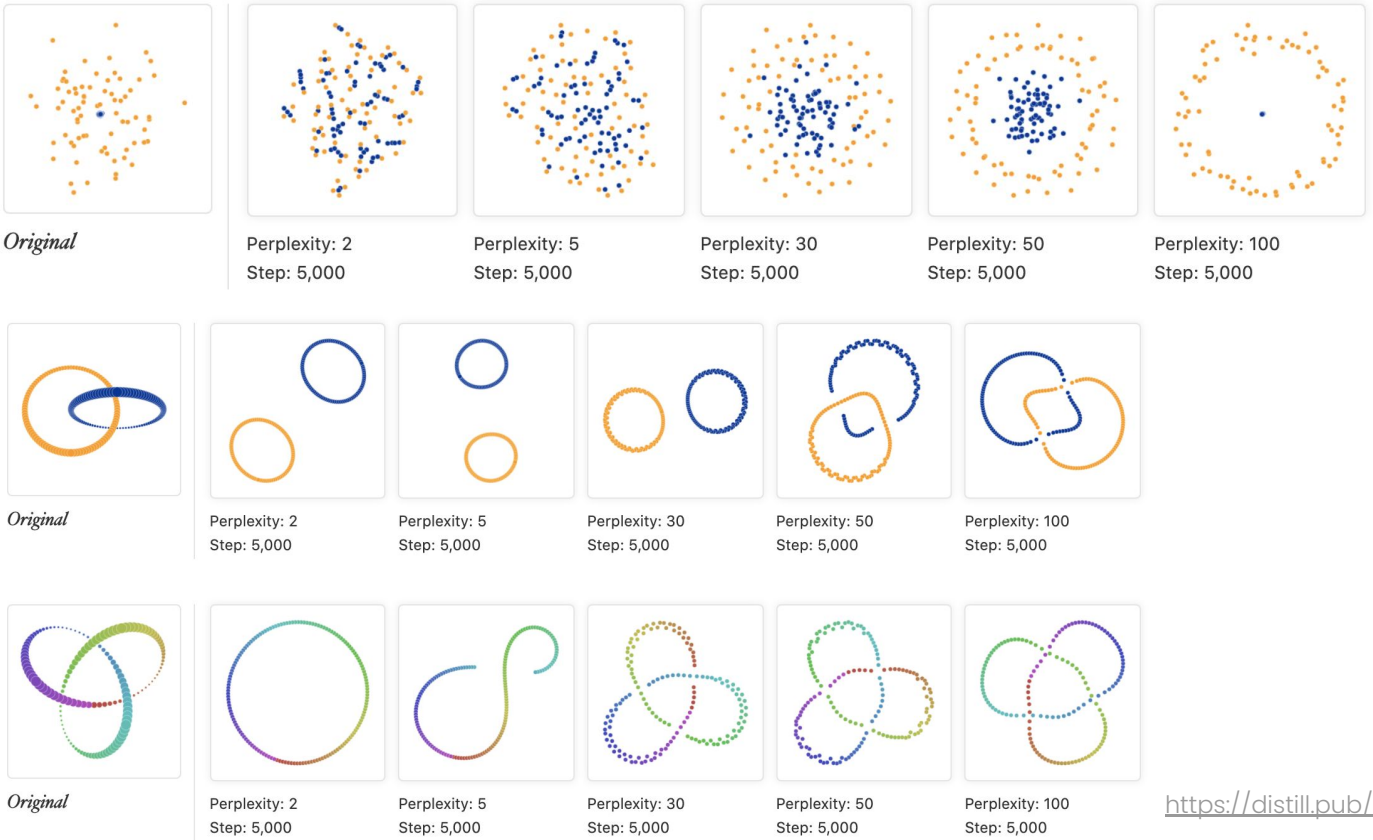
https://distill.pub/2016/misread-tsne/

5. You can see some shapes, sometimes



https://distill.pub/2016/misread-tsne/

6. For topology, you may need more than one plot



https://distill.pub/2016/misread-tsne/

Uniform Manifold Approximation and Projection (UMAP)

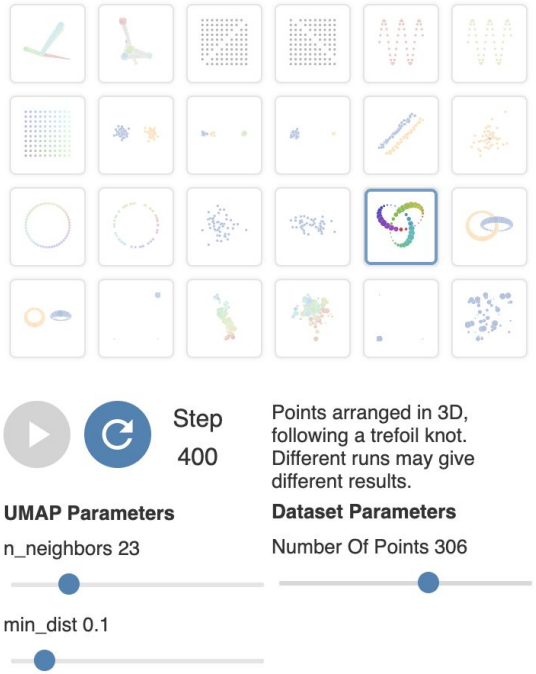
- Pros
  - Non-linear reduction that is computationally faster than t-SNE.
  - User defined parameter for preserving local or global structure.
  - Solid theoretical foundations in manifold learning.
- Cons
  - Requires setting hyperparameters that influence quality of the embedding.
  - Non-deterministic algorithm.



Article on Understanding UMAP

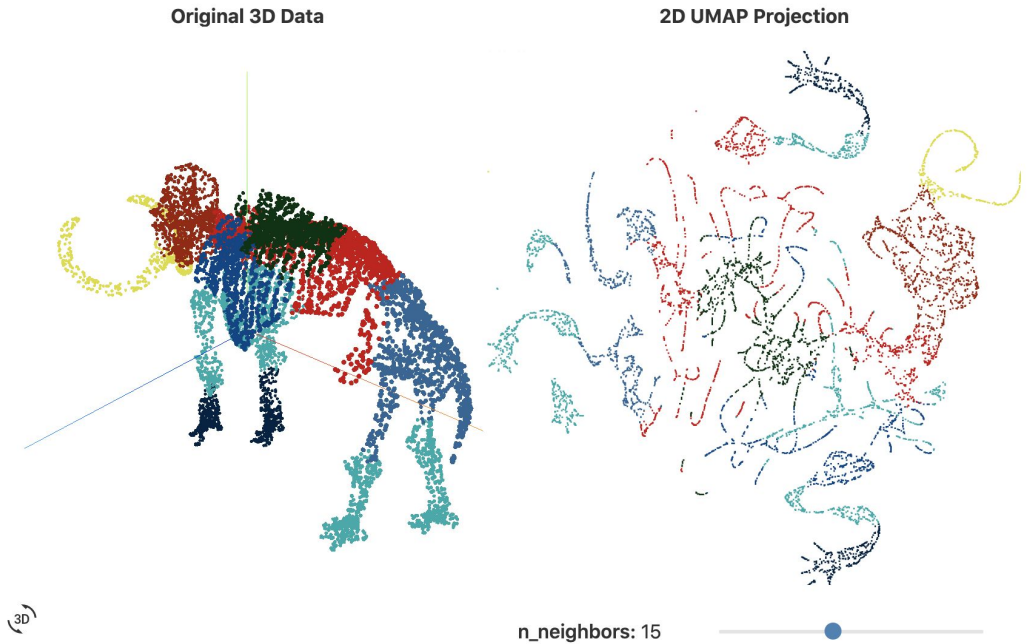
https://pair-code.github.io/understanding-umap/





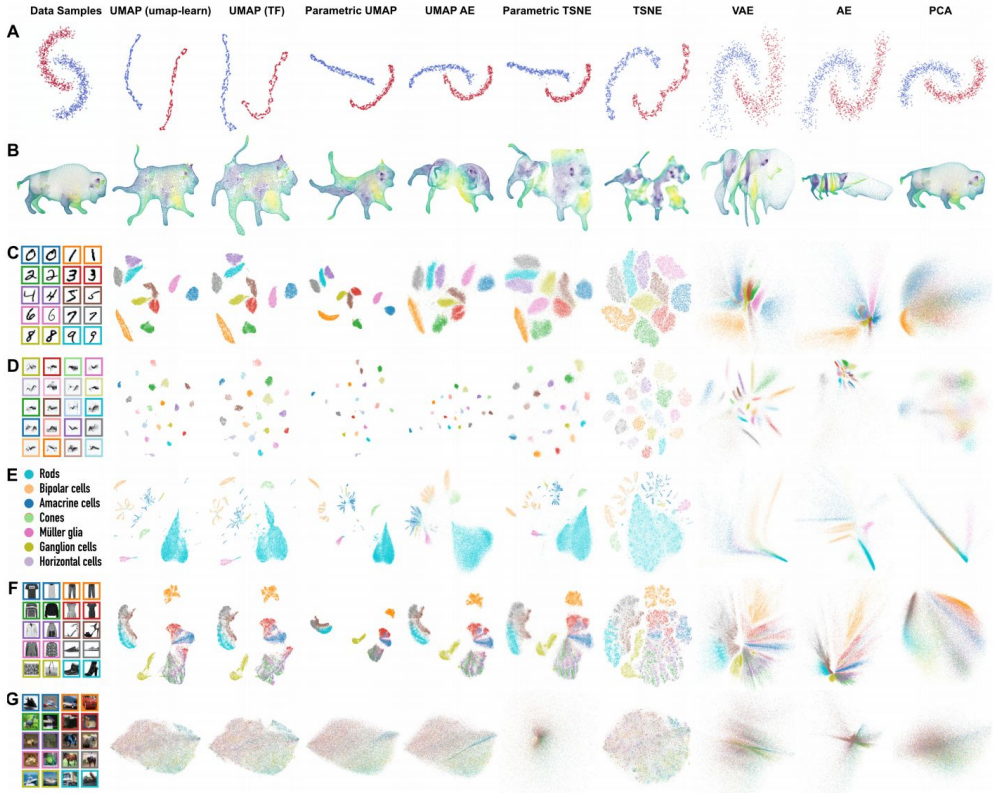
UMAP Projections of a 3D Woolly Mammoth Skeleton

50k points, 10k shown



https://pair-code.github.io/understanding-umap/

Comparison of Projection Techniques

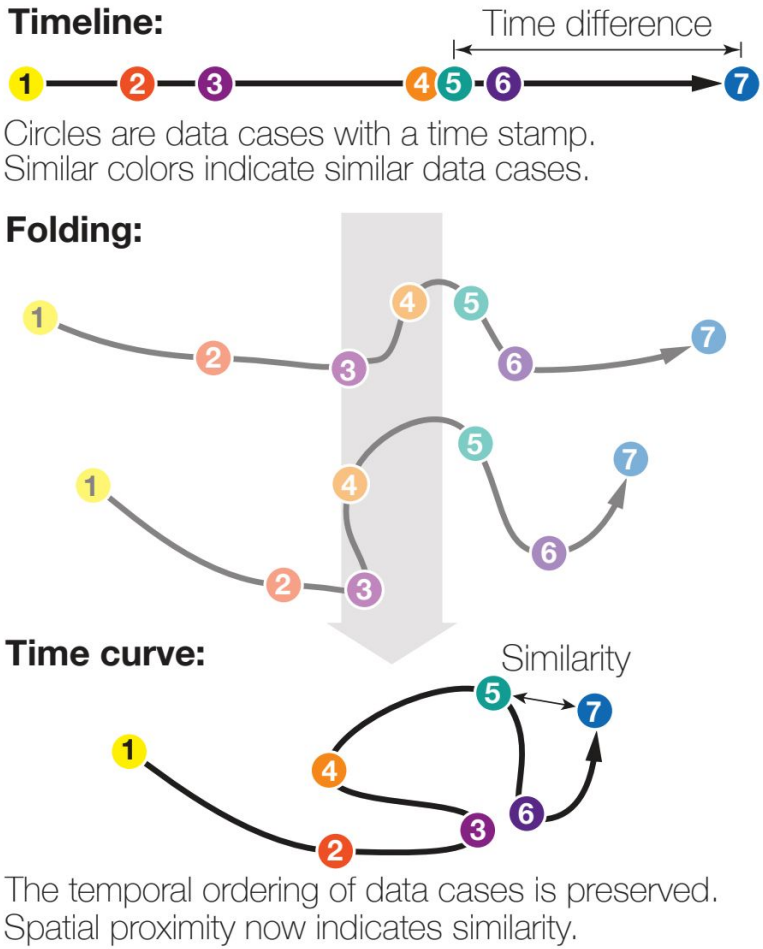


Sainburg et al. 2020

Embedding-based Trajectories

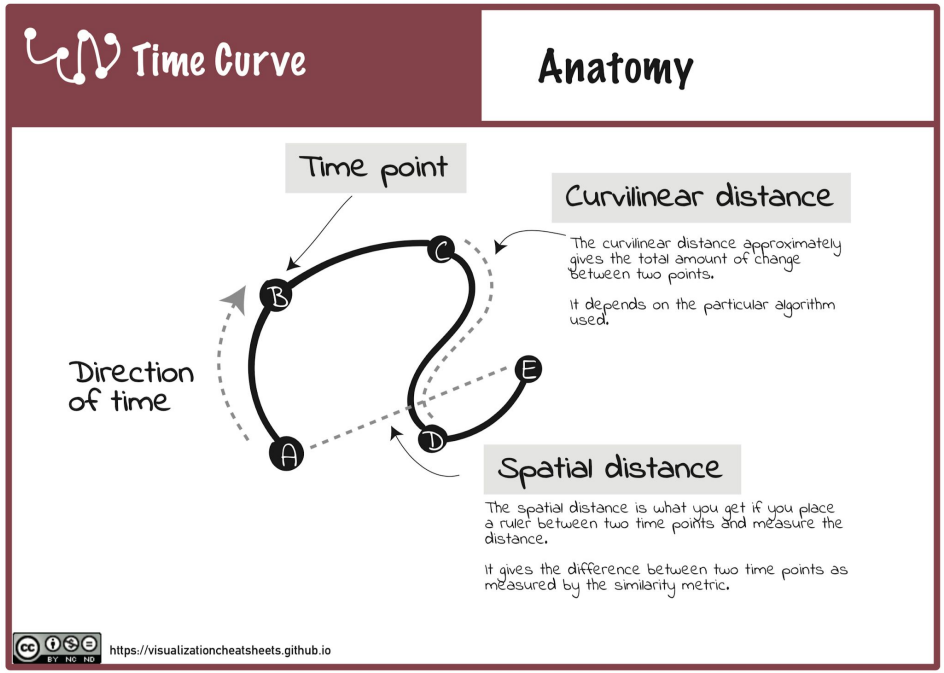
Time Curves Technique

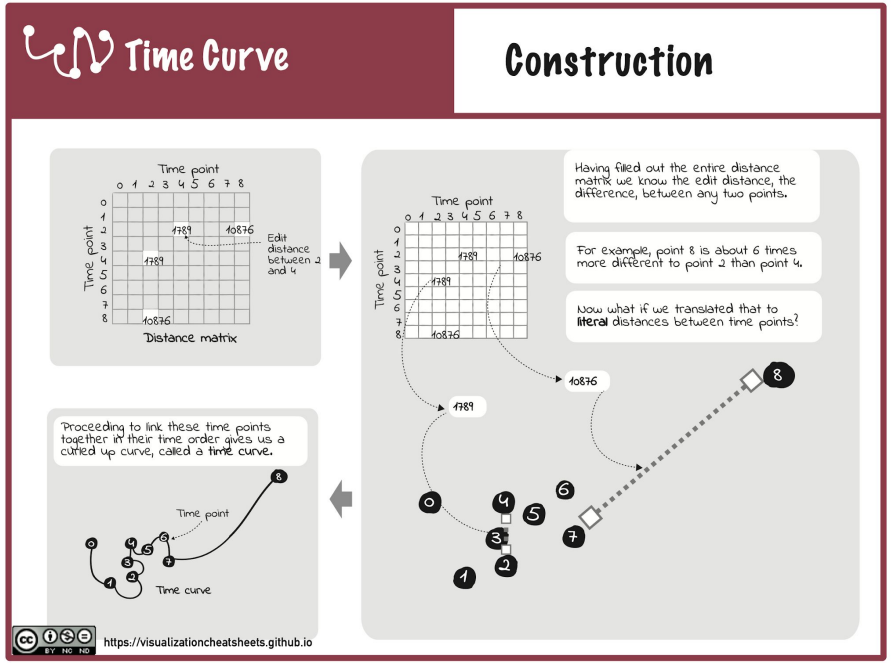
Folding time to visualize patterns of temporal evolution in data



Bach et al. 2015

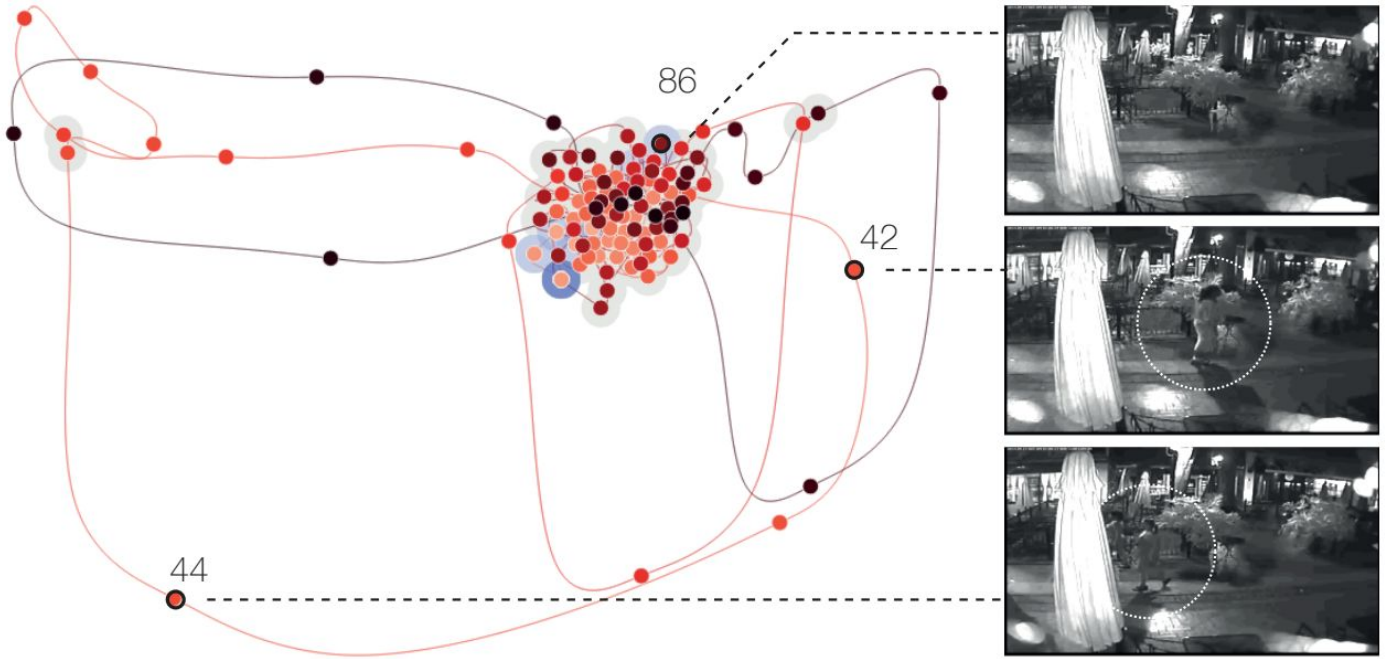
Cheat Sheet: Explanation of Time Curves Technique





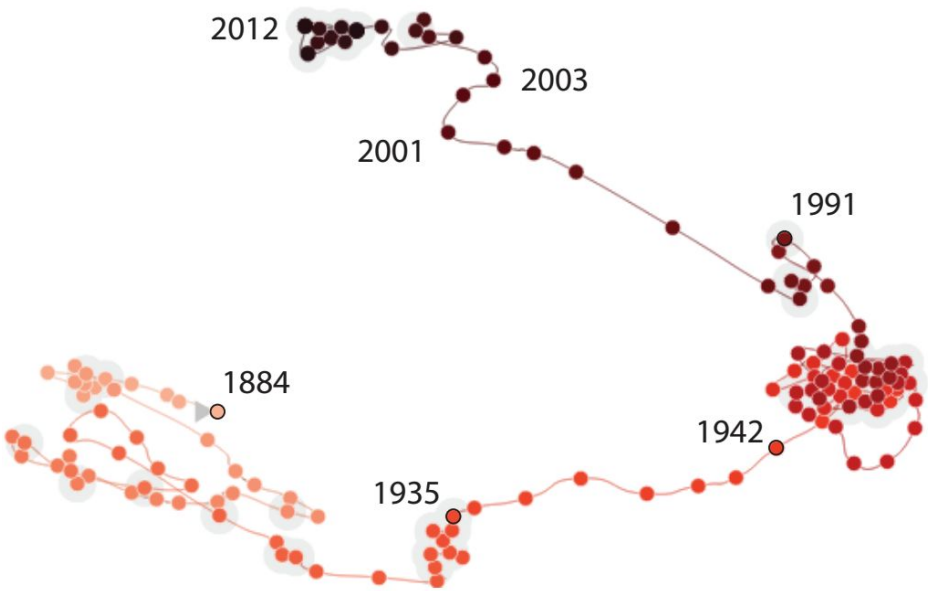
Time Curve Signature of Surveillance Video of a Street

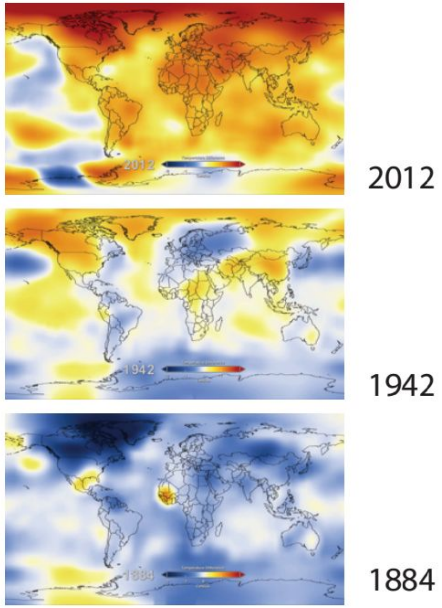
Outliers are passing pedestrians.



Bach et al. 2015

Time Curve Evolution of Global Temperature between 1884 and 2012

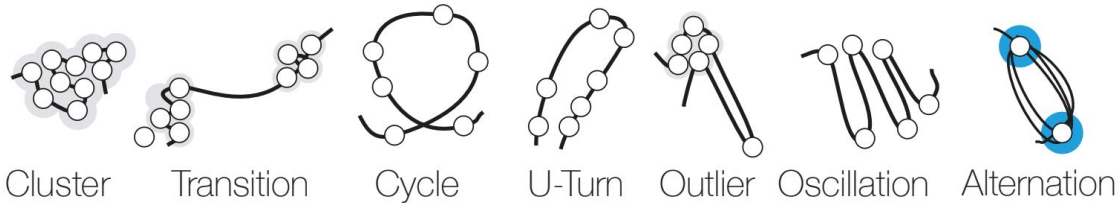




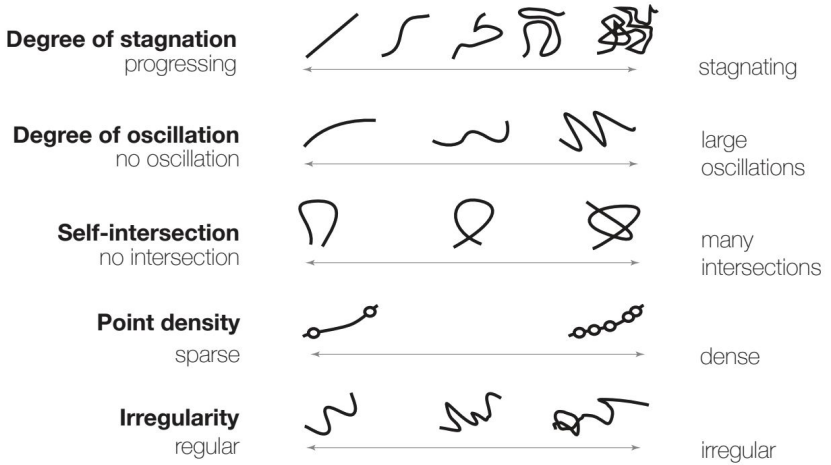
Bach et al. 2015

Time Curves Generalization of Patterns

Examples of visual patterns



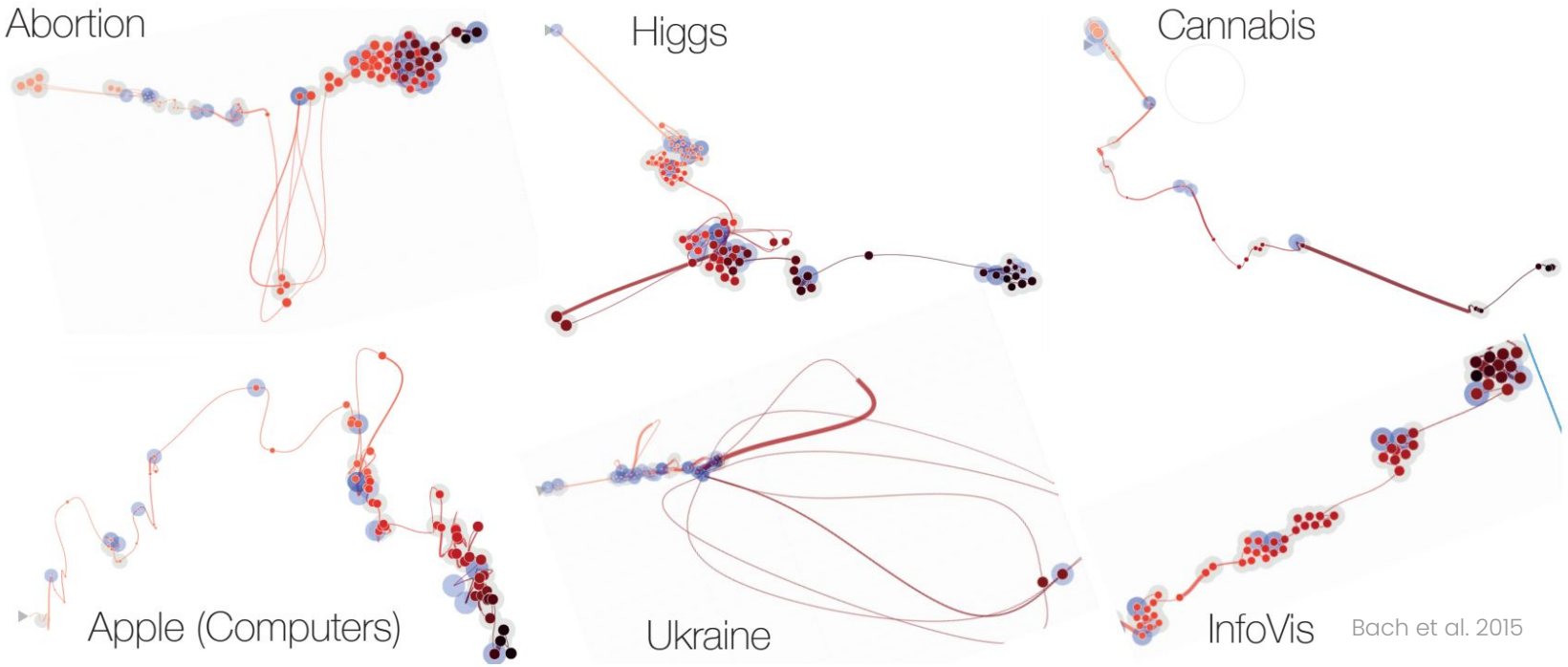
Five geometric characteristics



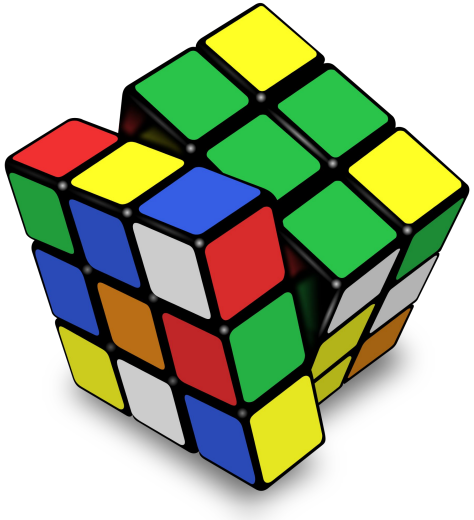
Bach et al. 2015

Time Curve Signatures of Different Wikipedia Articles

Loops are reverts. Oscillation are edit wars.



Bach et al. 2015



RUBIK'S CUBE



4.22s

HUMAN

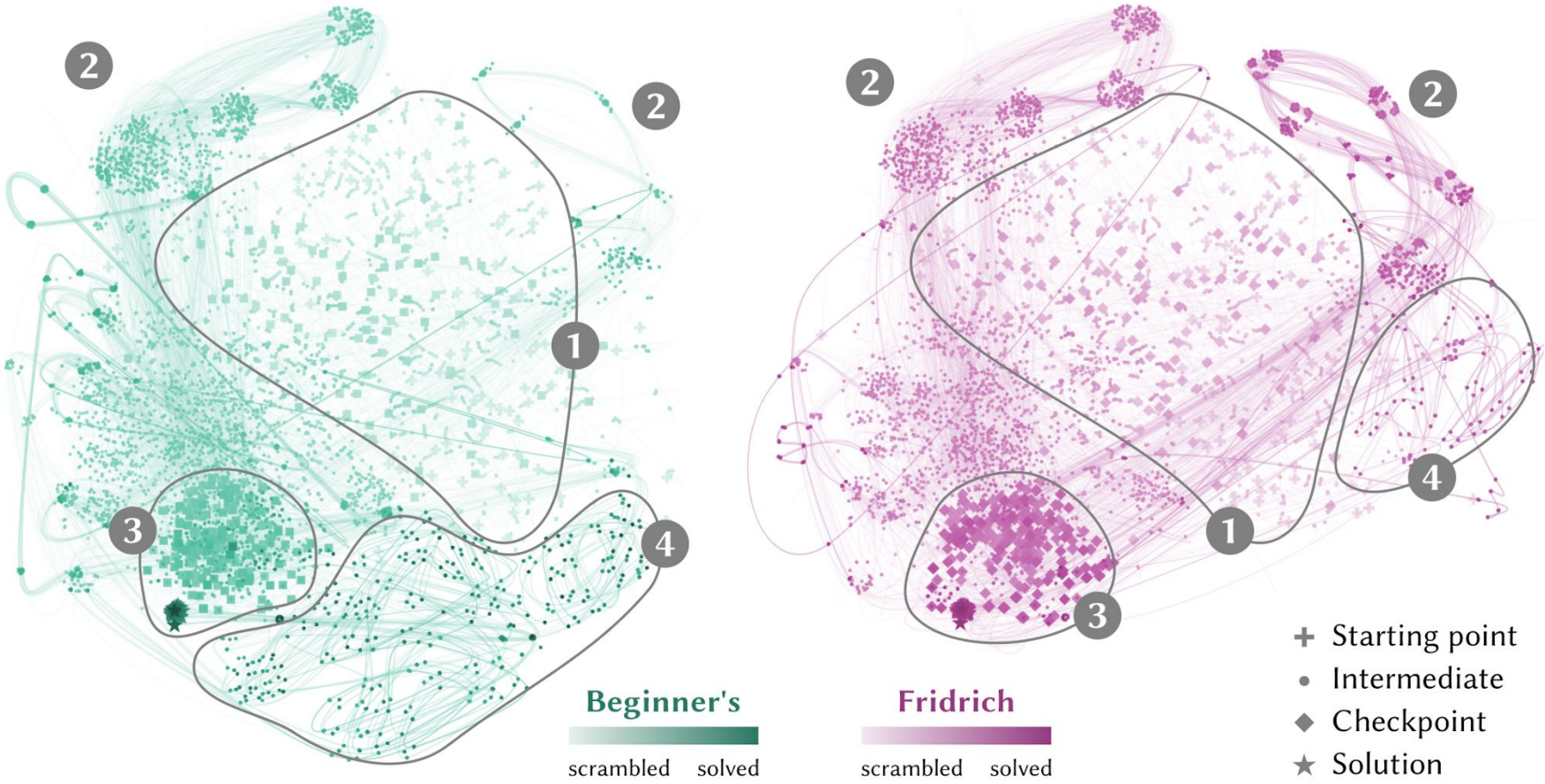
VS.

MACHINE



3
8

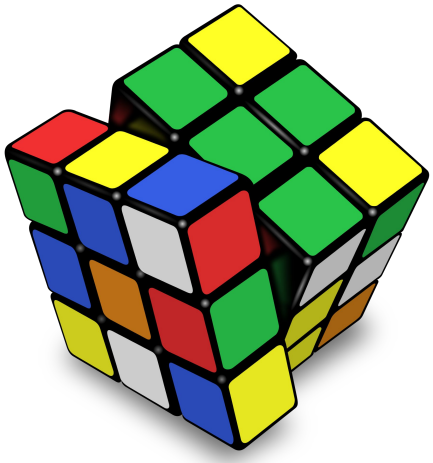
0.38s

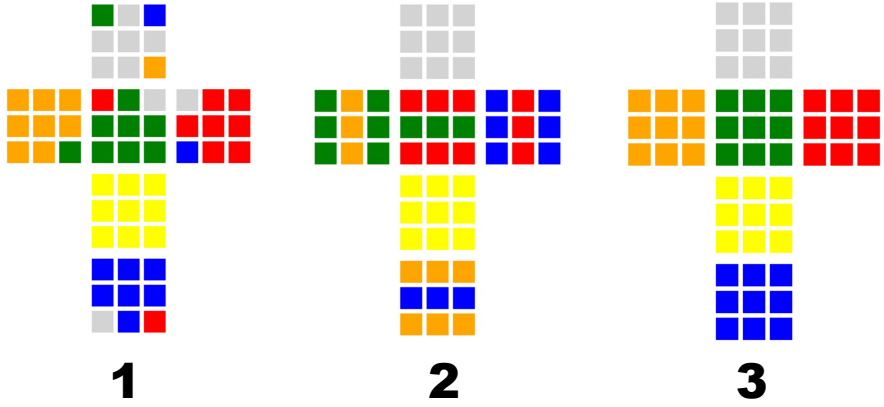


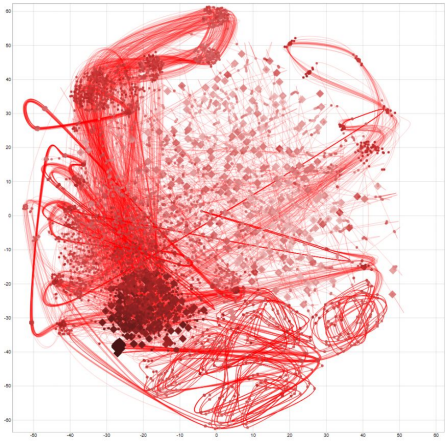
BEGINNER'S METHOD

FRIDRICH METHOD

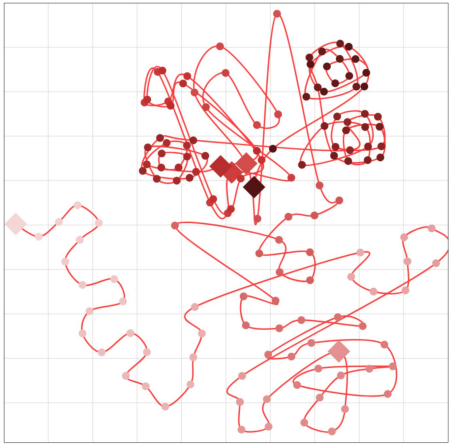
Steinparz et al. 2019



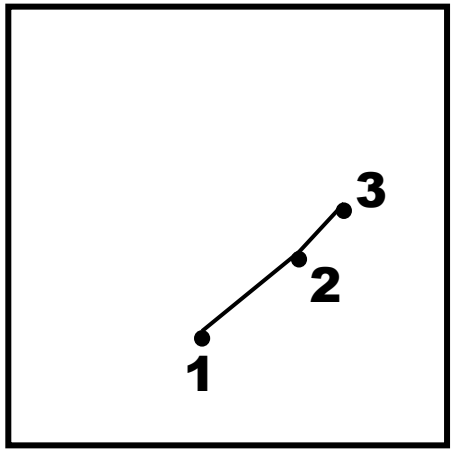




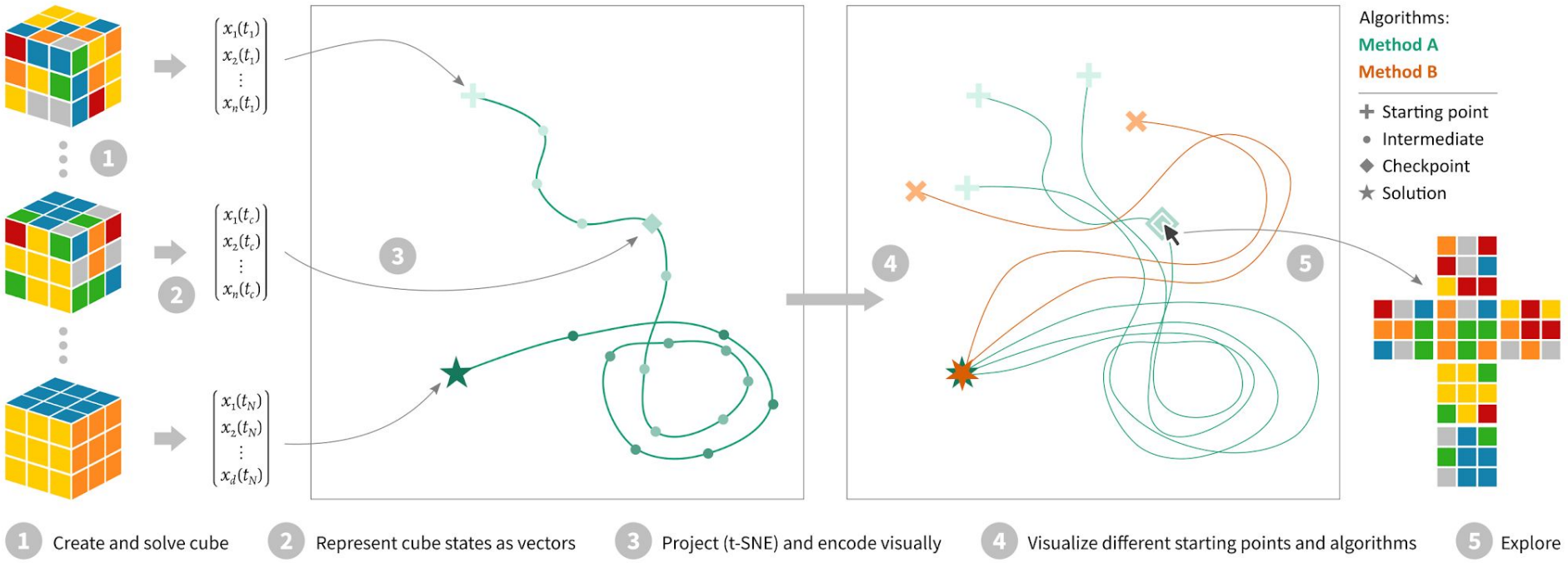
100 CUBES



1 CUBE



T-SNE PROJECTION



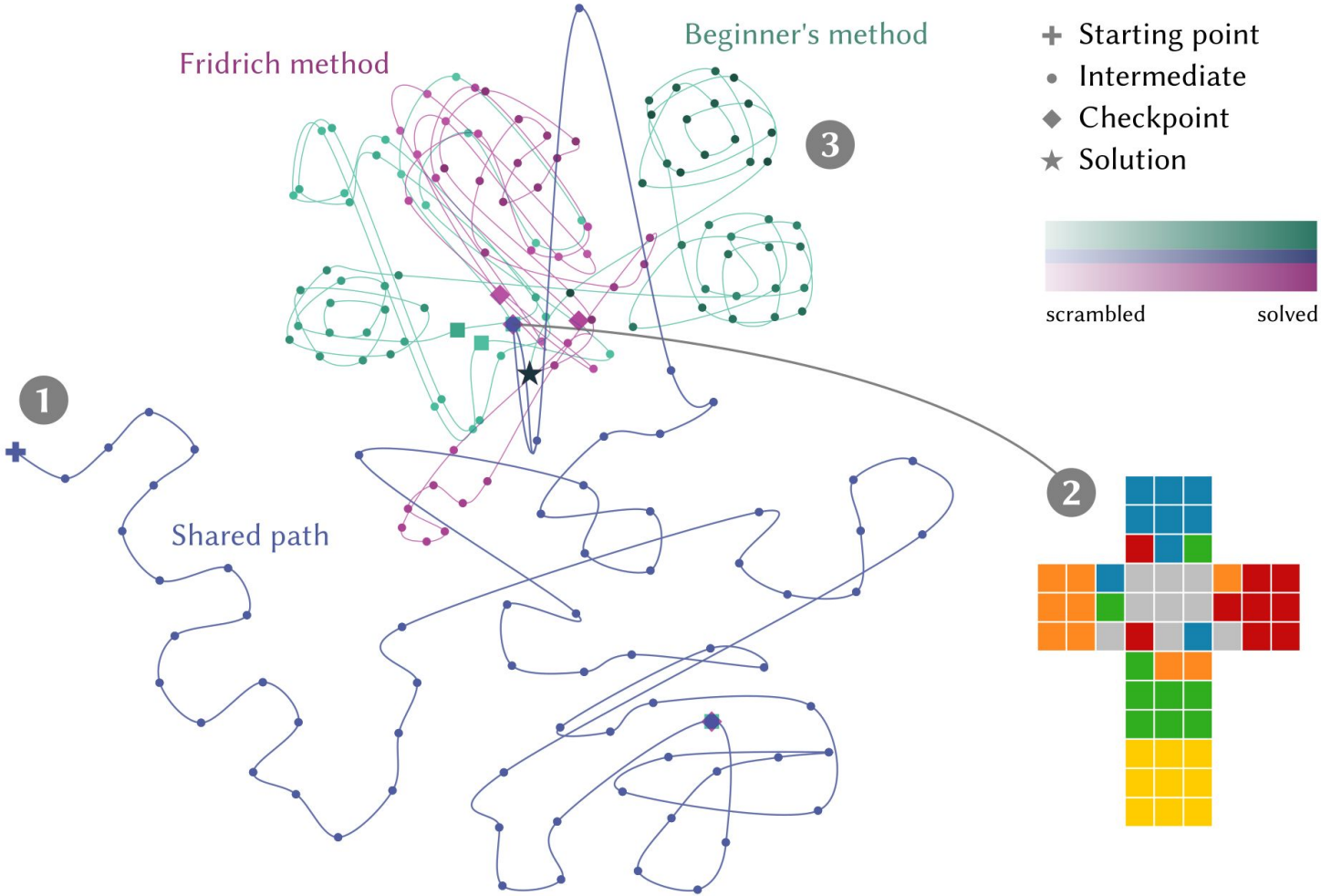
1 Create and solve cube

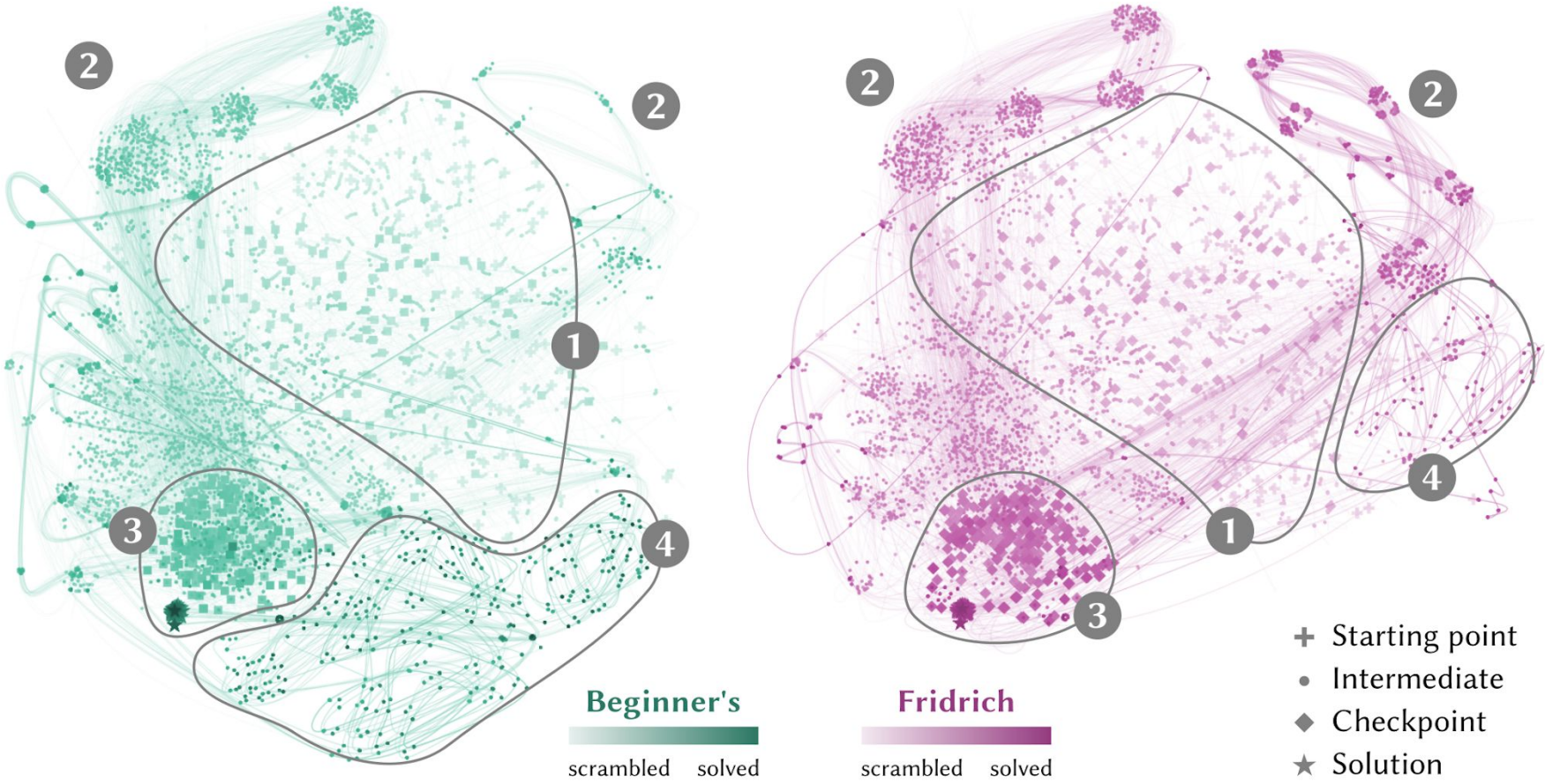
2 Represent cube states as vectors

3 Project (t-SNE) and encode visually

4 Visualize different starting points and algorithms

5 Explore


Fridrich method
Beginner's method
Starting point
Intermediate
Checkpoint
Solution
scrambled
solved
1
2
3
Shared path



BEGINNER'S METHOD

FRIDRICH METHOD

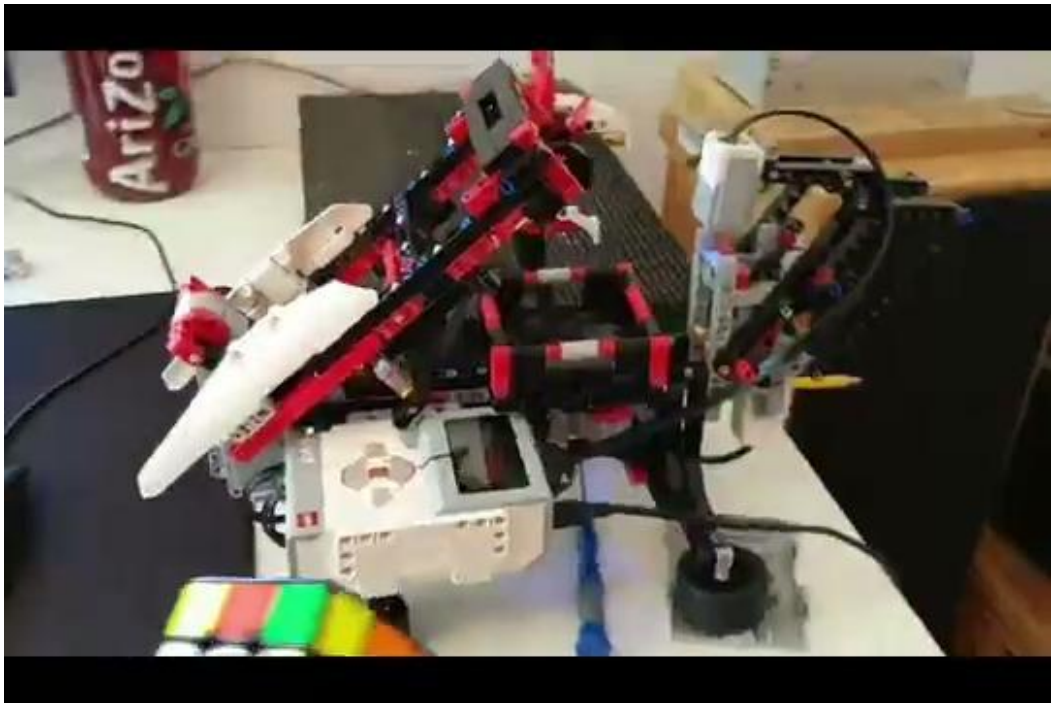
Steinparz et al. 2019

Projection Space Explorer Tool

https://jku-vds-lab.at/pse

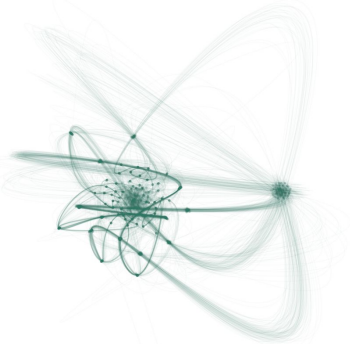


LEGO Mindstorm Solver





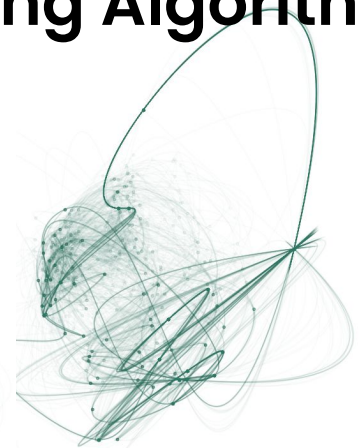
Application Example: Sorting Algorithms



UMAP Bubble Sort



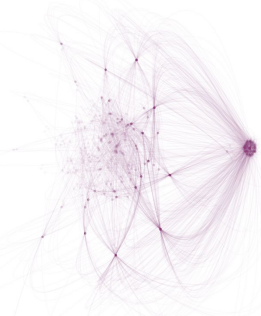
t-SNE Bubble Sort



Isomap Bubble Sort



UMAP Quick Sort

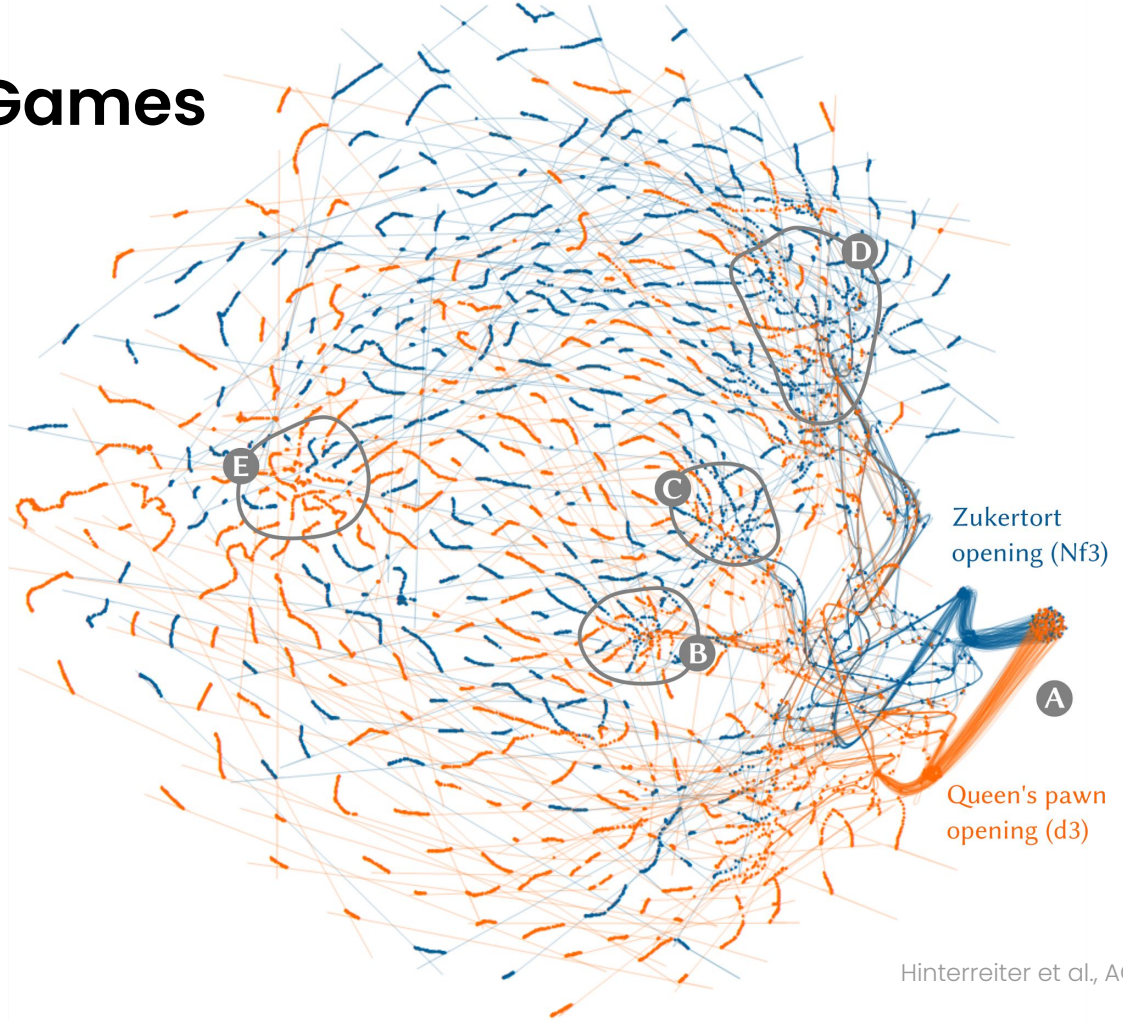


t-SNE Quick Sort

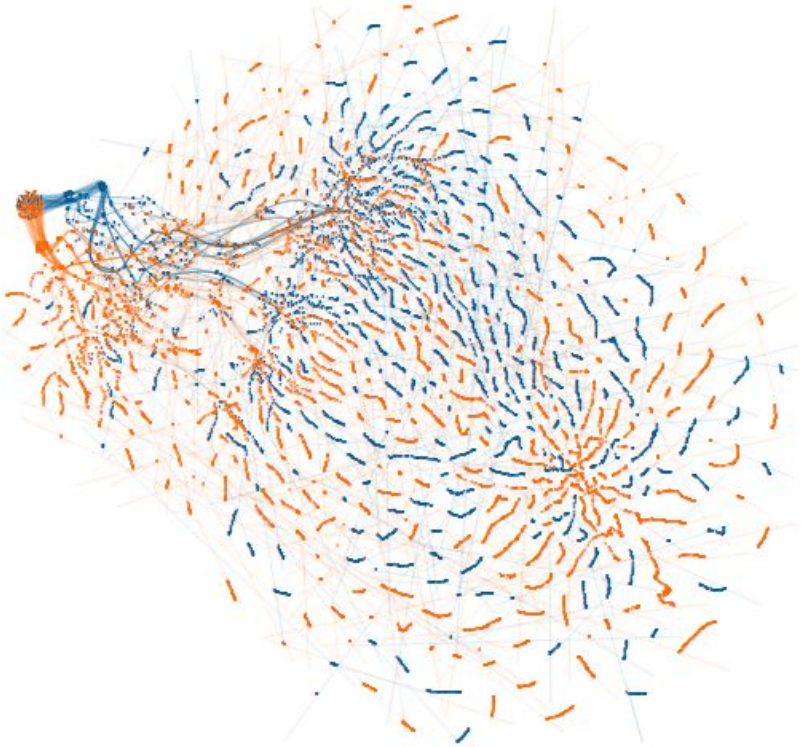


Isomap Quick Sort

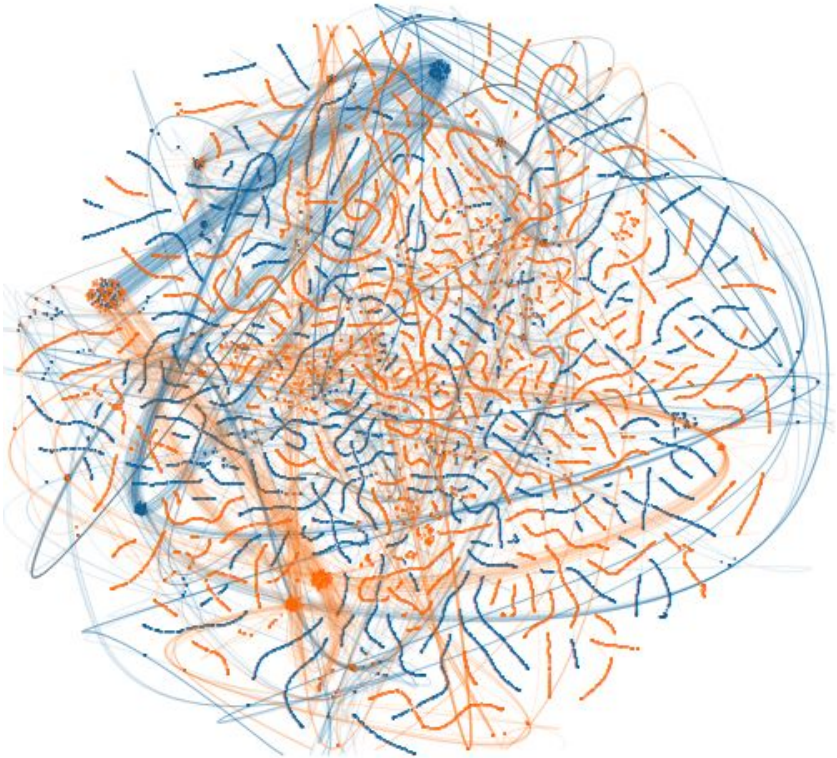
450 Chess Games



t-SNE Projection with Varying Perplexity



t-SNE with Perplexity 5

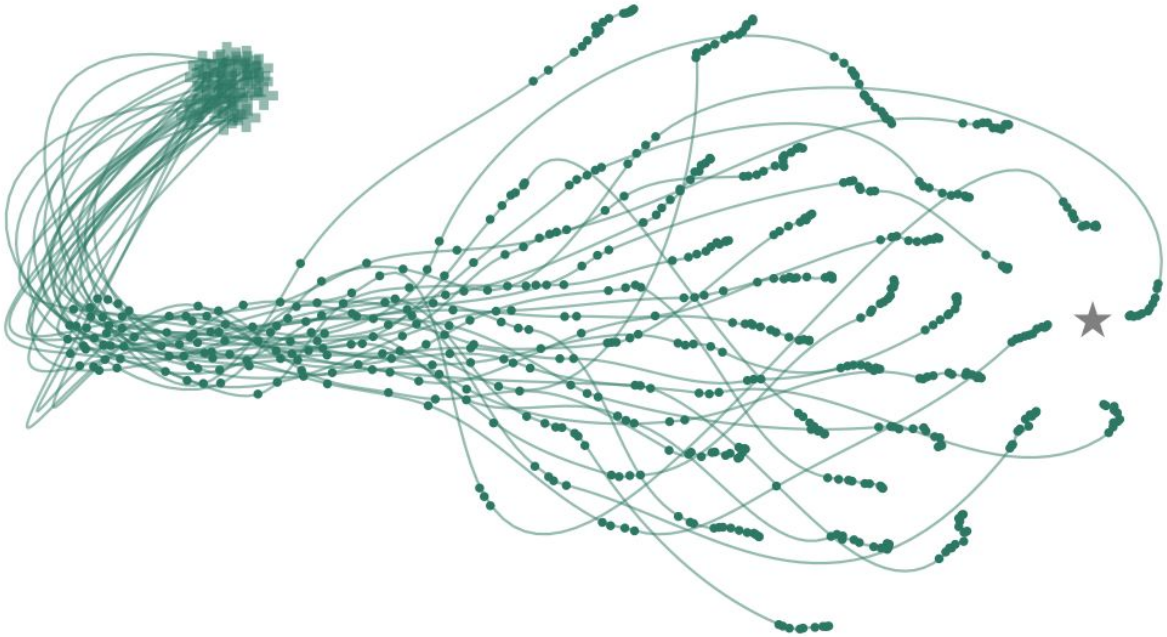


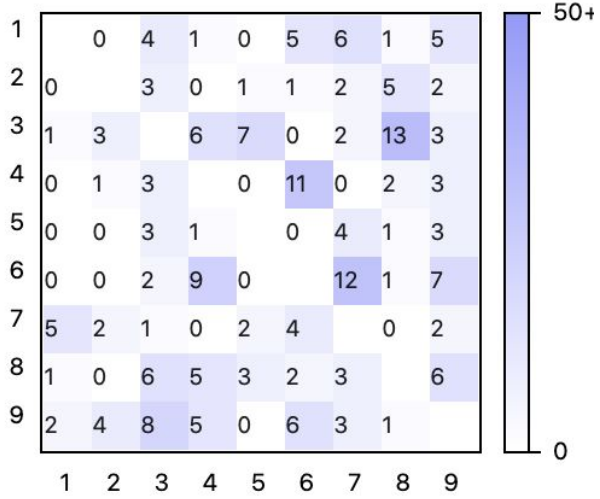
t-SNE with Perplexity 300

Projecting Neural Network Training

Distances between confusion matrices

Star shape = perfect classification





Hinterreiter et al. 2020

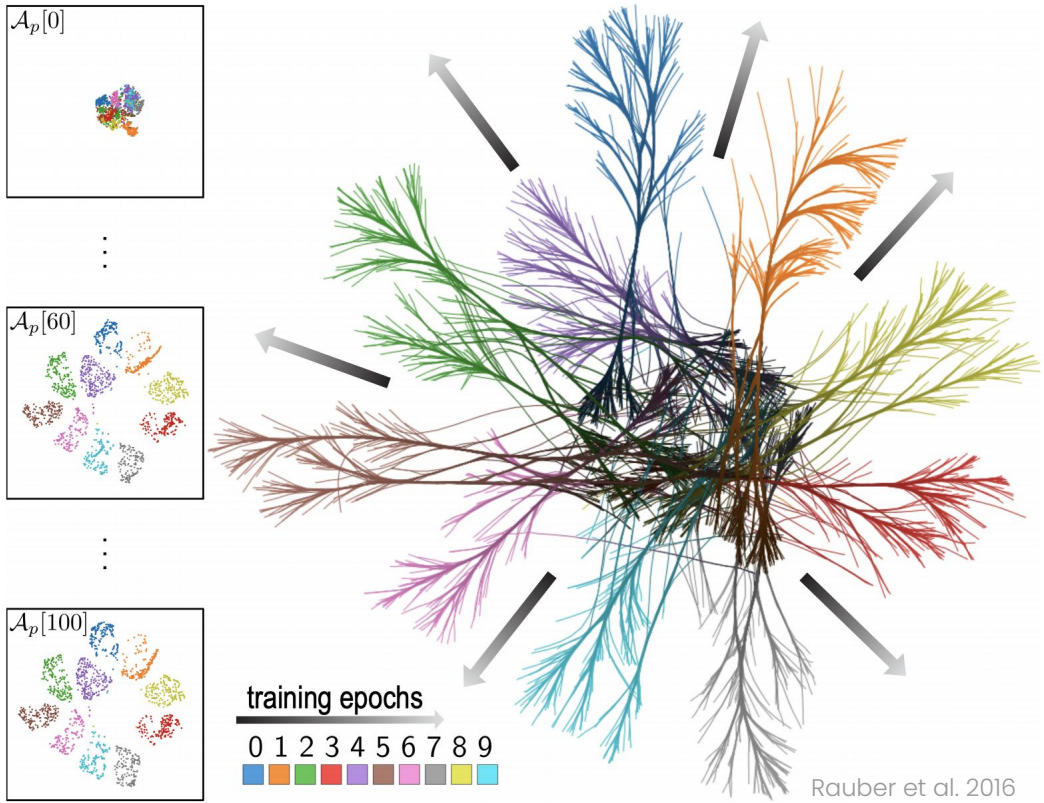
Inter-Epoch Evolution Using Trajectories

Last CNN hidden layer

Epochs 0-100, in steps of 20

MNIST test subset.

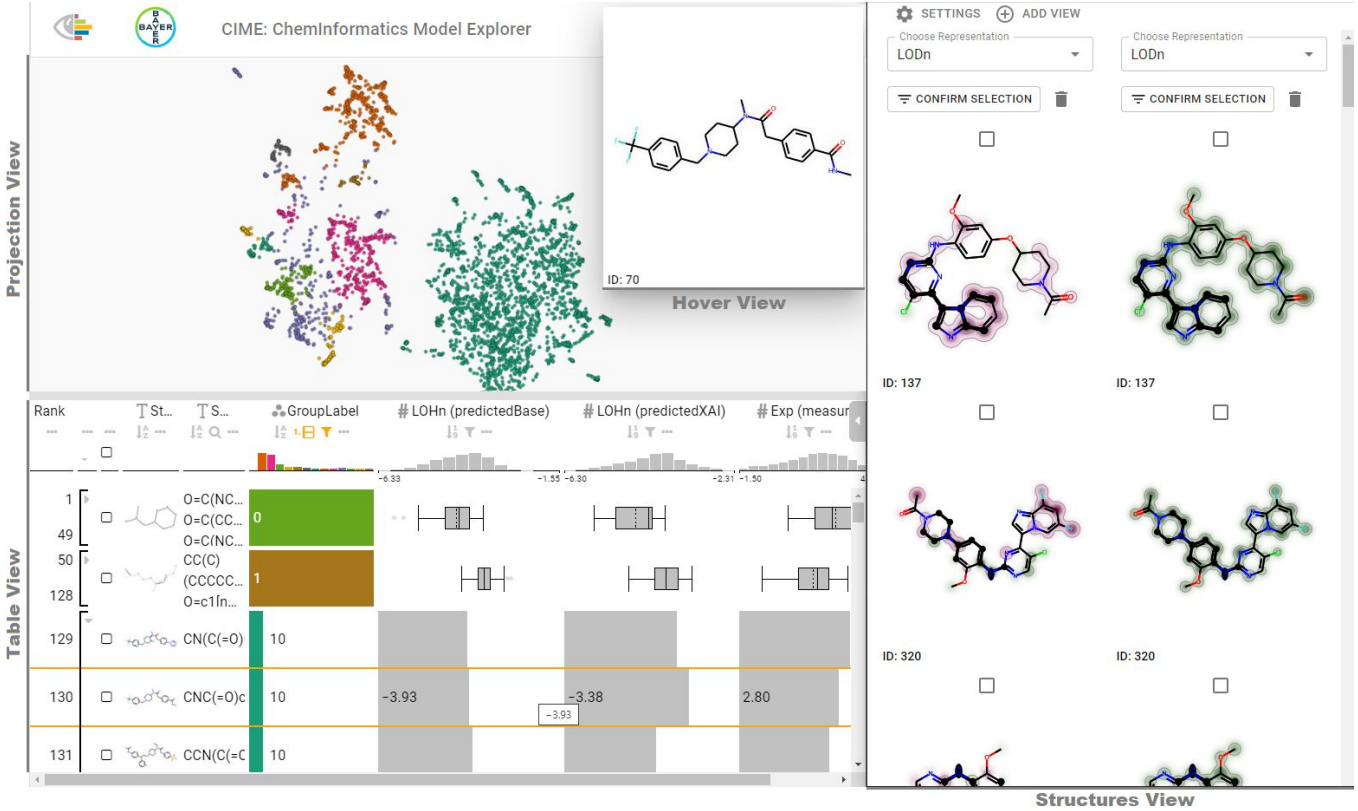
Brighter trail parts show later



Rauber et al. 2016

ChemInformatics Model Explorer (CIME): Exploratory Analysis of Chemical Model Explanations

datavisyn



Bayesian Reaction Optimization



Article

Bayesian reaction optimization as a tool for chemical synthesis

https://doi.org/10.1038/s41586-021-03213-y

Received: 24 June 2020

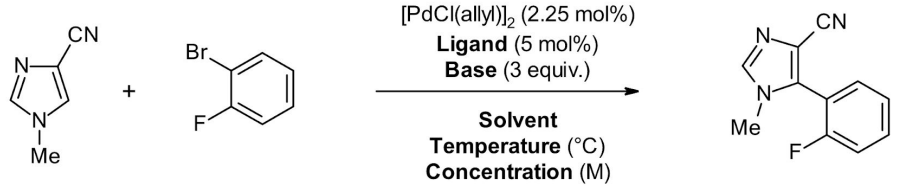
Accepted: 11 December 2020

Published online: 3 February 2021

Check for updates

Benjamin J. Shields[1], Jason Stevens[2], Jun Li[2], Marvin Parasram[1], Farhan Damani[3], Jesus I. Martinez Alvarado[1], Jacob M. Janey[2], Ryan P. Adams[3] ✉ & Abigail G. Doyle[1] ✉

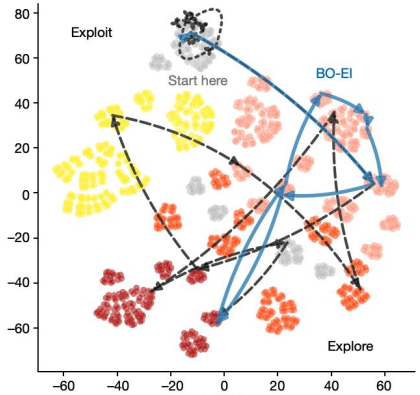
Reaction optimization is fundamental to synthetic chemistry, from optimizing the yield of industrial processes to selecting conditions for the preparation of medicinal candidates[1]. Likewise, parameter optimization is omnipresent in artificial intelligence, from tuning virtual personal assistants to training social media and product recommendation systems[2]. Owing to the high cost associated with carrying out experiments, scientists in both areas set numerous (hyper)parameter values by evaluating only a small subset of the possible configurations. Bayesian optimization, an iterative response surface-based global optimization algorithm, has demonstrated exceptional performance in the tuning of machine learning models[3]. Bayesian optimization has also been recently applied in chemistry[4–9]; however, its application and assessment for reaction optimization in synthetic chemistry has not been investigated. Here we report the development of a framework for Bayesian reaction optimization and an open-source software tool that allows chemists to easily integrate state-of-the-art optimization algorithms into their everyday laboratory practices. We collect a large benchmark dataset for a palladium-catalysed direct arylation reaction, perform a systematic study of Bayesian optimization compared to human decision-making in reaction optimization, and apply Bayesian optimization to two real-world optimization efforts (Mitsunobu and deoxyfluorination reactions). Benchmarking is accomplished via an online game that links the decisions made by expert chemists and engineers to real experiments run in the laboratory. Our findings demonstrate that Bayesian optimization outperforms human decisionmaking in both average optimization efficiency (number of experiments) and consistency (variance of outcome against initially available data). Overall, our studies suggest that adopting Bayesian optimization methods into everyday laboratory practices could facilitate more efficient synthesis of functional chemicals by enabling better-informed, data-driven decisions about which experiments to run.



Parameter Space:

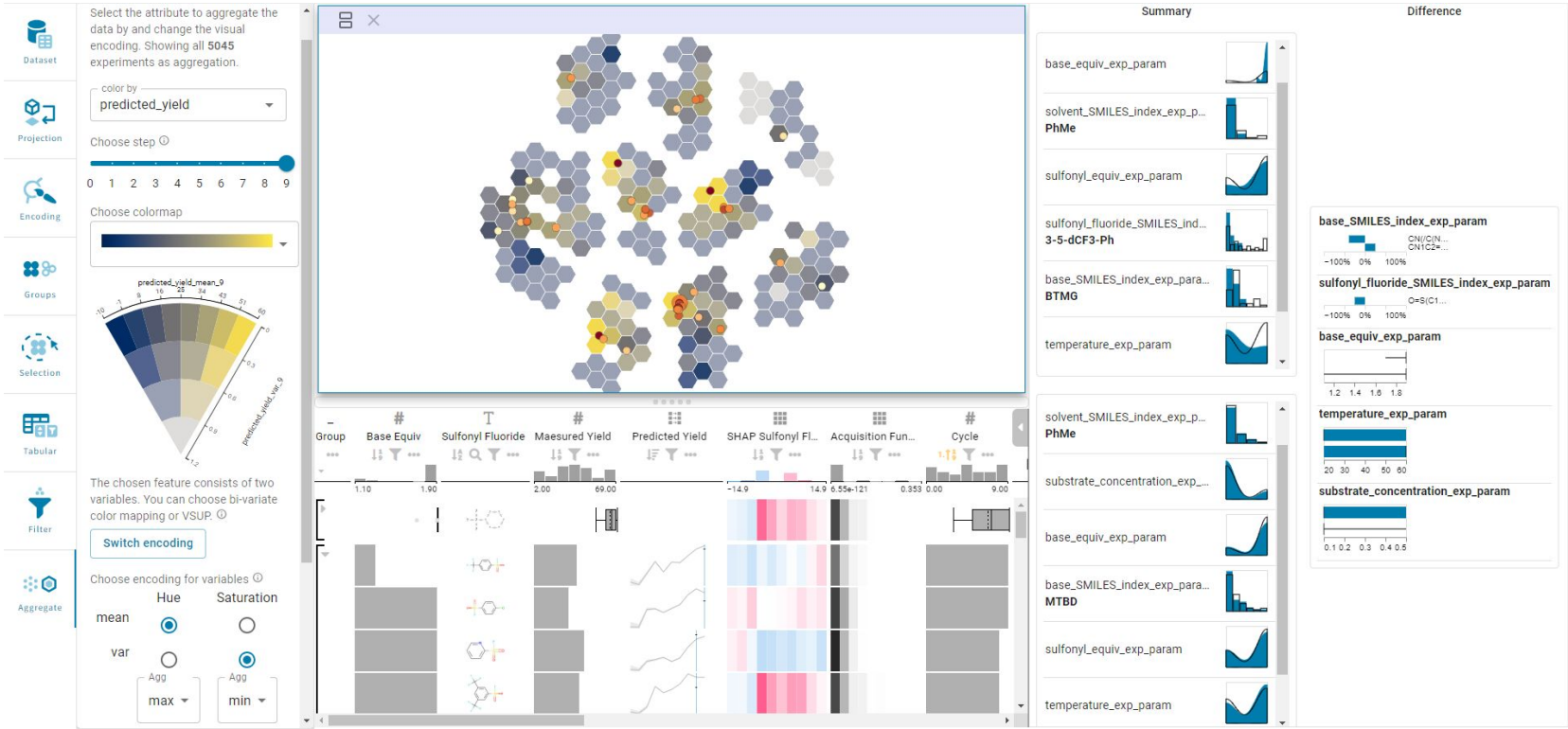
Categorical Variables:	Continuous Variables:
12 Ligands	3 Temperatures
4 Bases	3 Concentrations
4 Solvents	

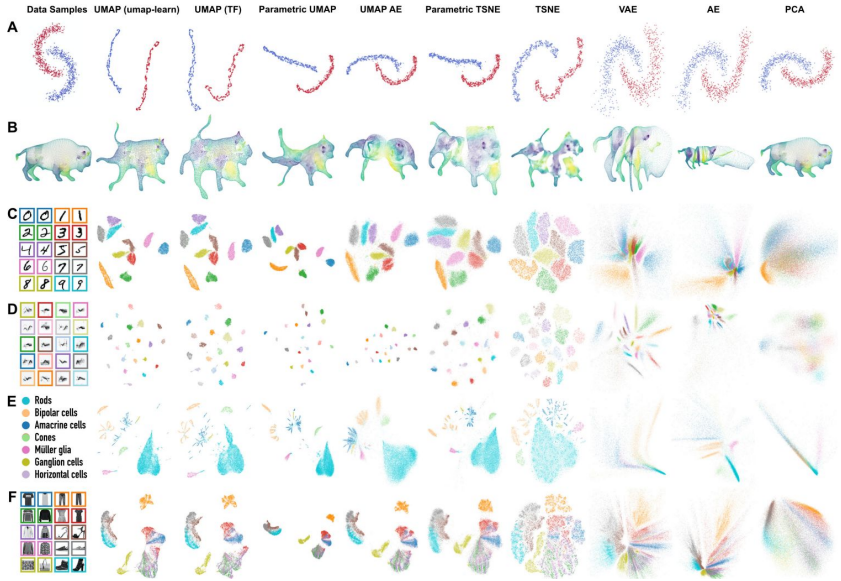




Shields et al. Nature 2021

CIME for Bayesian Reaction Optimization





Summary of Lecture 3

Explaining Through Projections

- **Dimensionality reduction techniques**
  - PCA, UMAP, t-SNE
  - Visualization of embeddings
- **Embedding-based trajectories**

Preview of Lecture 4

Visual Explanation Techniques I



vs



Intrinsic interpretability

Post-hoc explanation

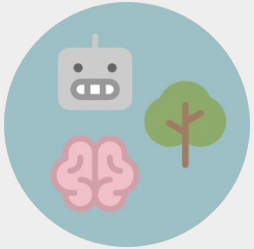


vs



Global

Local



vs



Model-agnostic

Model-specific



JKU
Visual Data
Science Lab

AI and Visualization

Univ.-Prof. Dr. Marc Streit

ai-and-vis-course@jku.at

Thank you!